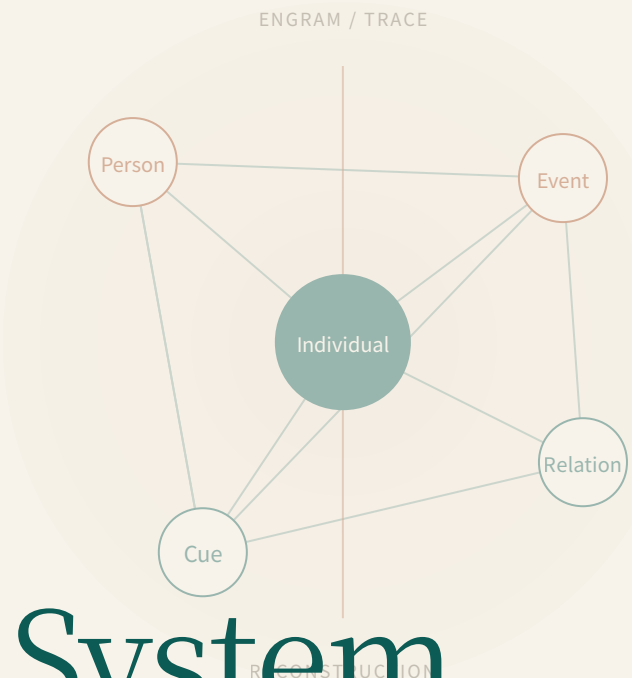


DUAL-BRAIN SYSTEM • VERSION 2.1

Dual-Brain System

A Multi-Model Collaborative Cognition Framework · From Neural Mechanisms to a Simulation Blueprint

> The primary purpose of this proposal is not to build a memory store, but to let multiple models divide the labor so that the context the Main Brain must read each round is compressed to a minimum. Memory is not a warehouse; it is the infrastructure that lets the Main Brain make better judgments within relationships. Encoding, consolidation, retrieval, reconsolidation, forgetting, decision-making, and social relationship memory are all links on this through-line.

Version 2.1 • Technical Proposal **Date** October 2026

Positioning A mechanistic deepening and engineering elaboration of Dual-Brain System, Version 1

10	8	6	4±1
BACKEND ROLES	MEMORY STORES	OPERATING LOOPS	CONTEXT SLOTS

FIG. 0 — THE RELATIONAL ENGRAM

CONTENTS

INTRODUCTION

00 The One-Page Proposal

PART I · MECHANISM

01 Memory Is Not Storage: A Dynamic System

02 From Neural Mechanisms to Computational Primitives

PART II · ARCHITECTURE

03 The Dual-Brain Architecture: One Main Brain, Ten Backends

3.6 Runtime Anatomy: From Permissions to Hard Boundaries

3.7 Ownership: Three Time Scales and the Ratification Right

04 Data Model: Events, Persons, Relationships, Interpretations

05 Six Operating Loops

PART III · SIMULATION

06 Simulation Details of the Key Mechanisms

07 Multi-Model Division of Labor and Context Economics

PART IV · VALIDATION AND DEPLOYMENT

08 Evaluation: Behave Like Memory, Not Like a Database

09 MVP and the Iteration Roadmap

10 Risks, Boundaries, and Open Questions

APPENDIX

A Glossary

B Key References

C Data Dictionary and Pseudocode

Suggested reading: start with 00 (the one-pager), 03 (architecture), and 09 (MVP) to build the overall judgment; then return to 01–02 (mechanisms) and 06 (algorithmic details); Chapter 08 provides the verifiable evaluation protocol.

00 • EXECUTIVE SUMMARY

The One-Page Proposal

The conclusion first: what must be simulated is not “memory”, but the loop between memory and decision.

The primary purpose of this proposal is not to build another memory store, but to let multiple models divide the labor so that the context the Main Brain must read each round is compressed to a minimum, while the information that does enter the context becomes more accurate. The human brain is not one large model; it is many functional units processing separately, with a single Main Brain making the unified judgment. The reason memory exists is to help the individual see clearly its relationships and position with others. Every mechanism below serves this through-line: **division of labor → context → judgment → relationship and memory update**.

THROUGH-LINE

Multi-model division of labor → the Backend compiles context → the Main Brain judges → relationship and memory update. Memory is not the goal; it is the infrastructure that lets the Main Brain make better judgments within relationships.

CORE THESES

The unit of memory is the Event, not the token; the reason memory exists is social relationships; the unit of context is the few blocks needed right now, not all of memory; the KPI is not “how much was stored” but “how many fewer errors in the next judgment within a relationship”.

FIRST-PRINCIPLES ABSTRACTION

Events affect state; state affects judgment and memory. Humans, machines, and large models share the same principle; they differ only in the carrier of state and the speed at which it updates. State must be split into `state_view` (enters the context, is the prompt) and `state_params` (enters the runtime, and genuinely changes retrieval, budget, gating, and writing).

> ARTICLE 0
>
> **Division of labor, not model stacking.** Multiple models each handle a small slice, and a single Main Brain makes the unified judgment; otherwise you have merely traded one large context for a crowd of small contexts.

Ten Design Principles

01 **Multiple systems, not multiple copies.** Episodic memory, semantic memory, procedural memory, working memory, and relationship memory are stored separately and updated separately; a single experience can produce three kinds of accumulation at once, but they share one Event source.

02 **Encoding has a threshold.** Not every input is worth remembering. Salience, goal relevance, social impact, emotional intensity, and prediction error jointly determine whether an engram is allocated.

03 **Index and content are separate.** The hippocampus does not store the whole film, only a sparse index; the system likewise does not stuff full text into the index, only pointers, sparse codes, and association weights.

04 **Consolidation is offline distillation.** Encode by day, replay by night: scan—cluster—replay—abstract—merge—prune, distilling episodes into semantics and skills.

05 **Retrieval is reconstruction.** Every answer is a narrative reassembled from fragments and must carry sources; where there is no source, abstain rather than fabricate.

06 **Reconsolidation is versioning.** Every retrieval may rewrite memory; new versions are appended, old versions are retained, and nothing is overwritten.

07 **Forgetting is a feature.** Decay, interference resolution, and index pruning jointly reduce noise; unfinished commitments and safety-critical information are exempt.

08 **Decision-making is dual-system.** When familiar, low-risk, and time-pressured, take the habitual fast path; when novel, high-risk, and requiring explanation, take model-based deliberation. Arbitration belongs to the Main Brain.

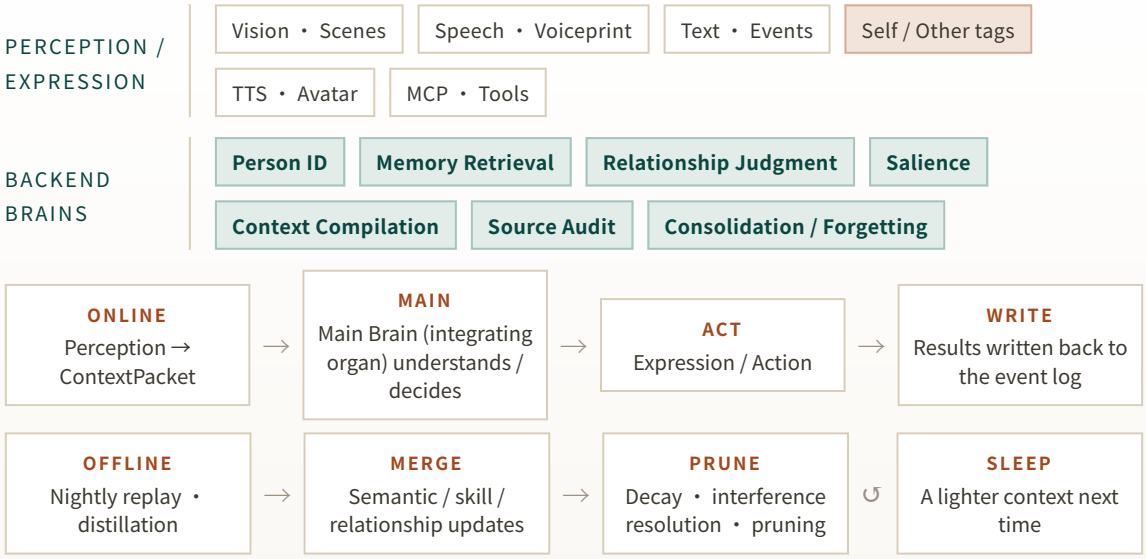
09 **Relationships are the master index of memory.** Person–Event evoke each other bidirectionally; relationships are multi-dimensional, asymmetric, evidence-bearing, and updated with time and experience.

10 **There is only one integrating center — and it belongs to the whole organism.** The Backend only compiles context: it does not offer advice, does not act, and does not decide in place of the Main Brain. The external acting identity belongs to the organism, and the Main Brain signs on its behalf within the authorized scope.

ARTICLES 11 AND 12 (IMPLICIT)

Boundaries are material, not policy; the ratification right is the ownership right. A constitution without runtime isolation is only a soft constraint; facts may be written by the Backend, but values/roles, goals, and habit rules are proposal-only, ratified by the Main Brain as the integrating organ, and the constitutional core requires equal co-signature by the user and the Main Brain. The execution mechanisms are in sections 3.6–3.7.

FIGURE 0 · THE ONE-PAGE ARCHITECTURE



Solid lines are the online loop, dashed lines the offline loop. The Backend may read the memory stores and submit write candidates, but it can influence the Main Brain only through the ContextPacket; the external acting identity belongs to the organism, and the Main Brain signs on its behalf within the authorized scope.

Relation to Versions One and Two

Versions One and Two established the skeleton of "one integrating center, backend brains, knowledge/memory/skill, relationships and experience, dual-brain decision-making". The current version keeps this skeleton but introduces three kinds of change: **retained and elaborated**, **revised**, and **added**. Revisions and additions must be marked, so that later discussion is not mistaken for the original formulation.

TABLE 0 · FROM VERSIONS ONE AND TWO TO THE CURRENT VERSION: RETAINED, REVISED, ADDED

ORIGINAL FORMULATION (V1/V2)	ELABORATION IN THE CURRENT VERSION	ENGINEERING LANDING POINT
The Main Brain has no context	The Main Brain is a stateless inference process, rebuilt each round from the ContextPacket; the Backend is a stateful memory system	ContextPacket as the sole cross-layer interface; the Main Brain does not persist conversation history

ORIGINAL FORMULATION (V1/V2)	ELABORATION IN THE CURRENT VERSION	ENGINEERING LANDING POINT
The Backend filters memory	Salience gating + hippocampal index + pattern separation/completion + context compilation	Encoding pipeline, retrieval pipeline, 4±1 slots, lazy expansion
Knowledge / memory / skill	Complementary Learning Systems: fast episodic store + slow semantic abstraction + procedural skill	Event log, semantic graph, skill library, sharing event_id
Relationships and experience	Person–Event bidirectional graph; relationships multi-dimensional, asymmetric, evidence-bearing	Three record types: Person / Participation / Relationship
Facts and interpretations kept separate	Source monitoring + interpretation version chain + evidence counted by event	Interpretation layer; `evidence_count = distinct event_id`
Memory fading	Ebbinghaus decay × relationship strength × future utility; unfinished commitments exempt	Retention formula, cold archive, index pruning
Habits like a cache	model-free fast path; caches only context templates and skills, never "decisions"	Habit table + Main Brain arbitration; the Backend does not exceed its authority
REVISION · Only the whole is the individual	The integrating center is an integrative property of the organism as a whole; the Main Brain is the integrating and signing organ, not the individual	3.1 / 3.6; the acting identity belongs to the organism, and the Main Brain signs on its behalf
REVISION · Habit rules require ratification	The Backend may propose by frequency, but a habit rule takes effect only with Main Brain ratification; the Backend does not directly rewrite the future self	3.7 / 6.7; Habit state candidate / ratified
ADDITION · Runtime boundaries	The Backend has no network, no tools, no persistent identity; the right to act belongs only to the organism	3.6; permission matrix, red-team checklist
ADDITION · Ratification ledger	Constitutive changes follow propose—ratify—cool down—roll back; the ledger is verifiable	3.7; Constitution / Ledger
ADDITION · Evidence ledger and simulation boundary	Recollection adds no evidence; conflicts are kept side by side; simulations are never written as facts	4.6 / 6.7.2; evidence_count / simulation schema
ADDITION · Compositional replay	Extract person/place/action/goal primitives from multiple events and compose future candidates by slot; flagged as simulation, never written as fact	6.2.1; primitives.py / simulation.py
ADDITION · Governance mode	Companion mode: the user and the Main Brain co-sign as equals; Engineering mode: the user holds the highest authority	3.7.1; ConstitutionStore(mode)

READING PATH

Main line: 0 → 1.9 → 3.1–3.3 → 4.2–4.4 → 5 → 6.8 → 7. Mechanistic background: 1–2, 6; engineering constraints: 3.6–3.7, 9.4, 10, which may be read later. This keeps “governance and safety” from being mistaken for the main line of memory.

PART I • MECHANISM

Understand the Brain First, Then Design the System

Memory is not a hard drive that can be read and written at will. It is a set of systems that compete with, inhibit, and complement one another: some handle fast binding, some slow abstraction; some maintain the present, some prune over the long term. To simulate it, you must first know what problem each part is solving.

01 Memory is not storage 02 From mechanism to computational primitives

Memory Is Not Storage: A Dynamic System

Encoding, consolidation, retrieval, reconsolidation, forgetting — five stages, five different computational problems.

If you compare memory to a library, you naturally think of "acquisition, indexing, borrowing". The brain is not like that. Not one book in the brain is preserved as it stands; every "borrowing" rebinds it, and may even rewrite a few pages. The brain is more like an archive that keeps rearranging itself: intake has a threshold, cataloguing relies on sparse indexes, a dedicated archivist merges similar files into summaries at night, and every reader leaves new marginalia. Understanding this process matters more than remembering the name of any brain region.

1.1 The Multi-System Map: Memory Is Not One Ability

Long-term memory divides into **declarative** (episodic: what one has experienced; semantic: what one knows) and **non-declarative** (procedural, priming, conditioning, habituation), plus a very small working memory and a relationship memory carrying emotional and social attributes. They depend on different brain systems, and their rules of writing and reading differ (Squire 2004; Eichenbaum 2004). The first design decision is not "build one larger store" but "build several stores with different write rates and read rules".

TABLE 1 · MEMORY SYSTEMS AT A GLANCE

SYSTEM	THE QUESTION IT ANSWERS	WRITE / READ CHARACTERISTICS
Episodic	When, where, and with whom one experienced what	Fast, sparse, single-shot write; depends on indexing and pattern separation
Semantic	What the world is, how concepts relate	Slow, overlapping, statistical learning; relies on repetition and sleep distillation
Procedural	How to do something	Gradual, automatized; error-driven and reinforced
Relationship	Who this person is, what we have been through	Person–Event bidirectional binding; multi-dimensional, asymmetric, evidence-bearing

1.2 Encoding: Salience Gating and the Hippocampal Index

Encoding first passes through attention and a salience gate: emotional arousal modulates hippocampal and cortical encoding (McGaugh 2000; Cahill & McGaugh 1998), and an unexpectedly good outcome is marked as worth learning (Schultz et al. 1997). The hippocampus does not store the whole experience, only providing a **sparse index** (Teyler & DiScenna 1986); the dentate gyrus performs **pattern separation** and CA3 performs **pattern**

completion. At the molecular level, the coincidence detection of NMDA, together with CaMKII and AMPA changes in LTP, requires transcription and translation to persist (Bliss & Lomo 1973; Malenka & Bear 2004; Davis & Squire 1984). A 2024 mouse study further found that CA1 engram cells show DNA double-strand breaks and TLR9–NF-κB signaling at encoding, suggesting that writing carries a cost and requires repair (Jovasevic et al. 2024; still a single mouse paradigm). Implication for the system: **a salience gate, an index gate, and an allocation gate** — missing any one of them and the design degrades into a full log.

1.3 Consolidation: Sleep Replay and Schema Assimilation

Synaptic consolidation stabilizes early LTP on the scale of minutes to hours (Frey & Morris 1997); **systems consolidation** hands the scaffolding over to the neocortex on the scale of days to years, while the hippocampus retains the index and detail (Squire & Alvarez 1995; Nadel & Moscovitch 1997). During sleep, slow oscillations, spindles, and hippocampal ripples couple, replaying the day's sequences (Staresina 2024; Schreiner et al. 2024); replay preferentially favors sequences that are highly salient, carry high prediction error, or relate to unfinished goals, and can also replay in reverse or simulate paths that never occurred (Foster & Wilson 2006; Pfeiffer & Foster 2013). Information compatible with a schema is assimilated faster (Tse et al. 2007). Implication for the system: **two paths — gist abstraction and detail retention**; first judge "which schema is this compatible with", then decide how deep to write.

1.4 Retrieval: Pattern Completion and Reconstruction

Retrieval is not playback but **reconstruction**: the current goal, emotion, and knowledge are used to assemble fragments into a self-consistent story (Bartlett 1932; Loftus & Palmer 1974). CA3 pattern completion lets a partial cue recall an entire memory, but it also over-completes: activating engram cells is sufficient to create a "false experience" (Ramirez et al. 2013). Encoding specificity determines whether a cue is effective (Tulving & Thomson 1973); **source monitoring** determines whether this was experienced, heard, or imagined (Johnson et al. 1993). Implication for the system: multi-dimensional cues (person, place, time, emotion, goal) plus source tags; retrieval without a source is a defense that does not exist.

1.5 Reconsolidation: Retrieval Is Rewriting

Retrieval pushes a memory back into an unstable state, after which it must be consolidated again (Nader et al. 2000). It is not that "every recollection necessarily rewrites": memory age, strength, prediction error, and contextual novelty determine whether updating occurs (Nader & Hardt 2009); the clinical use of propranolol to weaken fear memory remains at the research stage (Kindt et al. 2009). Implication for the system: **retrieval is writing, but write a new version, never overwrite the old one**.

> **The stability of memory is not "unchanged", but "safely updatable".**
>
> What reconsolidation teaches the system: retrieval is writing, but what gets written is a new version, not an overwrite of the old one.

1.6 Forgetting: Not a Failure, a Pruning

Ebbinghaus (1885) drew the first forgetting curve with nonsense syllables: memory drops quickly over the first hours to days, then flattens out. But "forgetting = decay" is only the shallowest explanation. The fuller picture is that forgetting arises from **decay, interference, retrieval failure, and active suppression** at the same time — and that it has adaptive value in its own right.

Richards & Frankland (2017) propose that forgetting is an "adaptive engram-cell plasticity": the memory trace persists but becomes hard to retrieve; this reduces interference of old memories with new learning and improves generalization. Mouse studies show that after natural forgetting the original engram cells can still be reactivated, so the memory is not entirely gone (O'Leary et al. 2024). Other studies reveal more "active" mechanisms: microglia can clear forgetting-related connections through complement-dependent synaptic pruning (Wang et al. 2020); adult hippocampal neurogenesis also promotes forgetting of old episodic memories (Akers et al. 2014).

Interference is another major source of forgetting. Two similar bodies of material learned in succession compete: the earlier one is disrupted by the later (proactive interference), and the later by the earlier (retroactive interference). The brain uses pattern separation to reduce this competition; when separation fails, you get the experience of "mixing things up." For a system, **forgetting is not deleting the log but lowering index weights, cutting wrong associations, and moving low-value content into cold archive**. The raw events should be retained for audit and traceability; what gets pruned is "current retrievability."

> DESIGN IMPLICATION
>
> A system that never forgets is crushed by old information: retrieval slows down, interference grows, the context gets dirty. Forgetting must be a first-class citizen with **a formula, exemptions, and auditability**, not "delete once storage is full."

1.7 Working Memory and Executive Control: The 4±1 Bottleneck

Working memory is "what you are thinking about right now." Miller's (1956) "7±2" is widely cited, but the more accurate estimate comes from Cowan (2001): without rehearsal and chunking, an adult can hold about **4±1** independent chunks simultaneously. Baddeley and Hitch (1974) split working memory into a central executive, a phonological loop, and a visuospatial sketchpad, later adding an episodic buffer (Baddeley 2000). These components are not anatomically separate boxes but a functional division of labor.

Prefrontal neural activity appears both as sustained firing (Fuster & Alexander 1971; Goldman-Rakic 1995) and as an "activity-silent" hidden state: information is temporarily absent from firing but can be reactivated via synaptic weights or network state (Stokes 2015). This gives a system a key insight: **the context does not have to stay "lit."** Working-memory slots can hold only pointers and lazily expand from long-term memory when needed; this is closer to what the brain does than stuffing every detail into the prompt.

Working-memory updating is controlled by **gating**. Basal ganglia–prefrontal loops decide when to "open the gate" to let new information in, when to maintain the current goal, and when to inhibit interference (Frank et al. 2001; Miller & Cohen 2001). Open the gate too often and you are constantly interrupted; too seldom and you become stuck. A multi-model system needs such gating too: not every new piece of information deserves entry into the Main Brain's context.

1.8 Value, Emotion, and Decision: Memory Serves Action

Memory does not exist to recollect for its own sake. What it must answer is: **what to do next**. This ties the memory system to the value system, the emotion system, and the decision system.

The amygdala tags events with "emotional salience," especially threat and reward; through its interactions with the hippocampus and prefrontal cortex it determines which experiences are worth encoding and consolidating preferentially (LeDoux 2000; Phelps 2004). The ventromedial prefrontal cortex and orbitofrontal cortex integrate multidimensional value, converting "this person, this event, this outcome" into subjective utility (Kable & Glimcher 2007; Hare et al. 2009). The dorsal striatum supports **model-free** habitual decisions: repeated "situation–action–outcome" sequences get compressed into a fast pathway, while the ventromedial prefrontal cortex and hippocampus support **model-based** deliberation, simulating the consequences of different choices in the mind (Daw et al. 2005; Dolan & Dayan 2013).

"Reward prediction error" (RPE) is the core signal of this learning system: when the outcome is better than expected, reinforce the preceding step; when worse, weaken it (Schultz et al. 1997; Montague et al. 1996). This maps directly onto a formula for updating the system's habits and values: `δ = r − V(s)`. Damasio's "somatic marker" hypothesis reminds us that emotion is not noise in decision-making but part of value integration (Damasio 1994; Bechara et al. 1994).

Stress alters this system. Acute stress generally promotes consolidation of emotional memories while impairing prefrontal-dependent retrieval and executive function (de Quervain et al. 2000; Cahill 2003). This explains a common phenomenon: the more nervous you are, the harder it is to recall a familiar name. A system need not deliberately "simulate stress," but it should have a "high-load/high-risk" mode: raise verification thresholds, reduce diffuse association, prefer the most reliable sources.

Finally, **episodic future thinking**: the hippocampus not only reconstructs the past but also participates in constructing future scenarios. Patients with hippocampal damage are equally impaired at imagining novel, complex future situations (Hassabis et al. 2007); the default mode network overlaps heavily between remembering the past and imagining the future (Schacter et al. 2007). This gives a system a direct capability: **use memory fragments to run "future simulations,"** not merely to answer "what happened in the past."

1.9 Social Relationship Memory and the Self: Memory's Other Main Axis

Why do humans need memory? Version one gave a good answer: because humans are social animals, and memory helps us understand our relationships and standing with others. This answer has a neuroscientific counterpart. Person knowledge, theory of mind (inferring others' beliefs and intentions), and autobiographical memory share a network comprising the medial prefrontal cortex, temporoparietal junction, posterior cingulate, and angular gyrus (Frith & Frith 2006; Saxe & Kanwisher 2003; Buckner & Carroll 2007). The default mode network is active when "thinking about oneself, about others, about the past, about the future," indicating that self, other, and time are different facets of the same simulation system (Spreng et al. 2009).

Autobiographical memory is not a diary ordered by time; it is jointly constructed by an "autobiographical knowledge base" and a "working self" (Conway & Pleydell-Pearce 2000). What we remember are the fragments relevant to our current self-goals, values, and relationship states; narrative identity strings these fragments into a coherent life story (McAdams 2001). This means: **relationship memory is not an appendage of episodic memory but a main axis that co-evolves with the self-model**. An experience matters often not because it carries "a lot of information" but because it changed "who the individual is and what relationship the individual has with whom."

A caveat is needed: relationship memory has no separate "relationship brain region." It is a distributed combination of person knowledge, emotional appraisal, theory of mind, and episodic memory. For system design, making it an independent **person–event–relationship graph** is a reasonable engineering abstraction, but one should not claim it corresponds to a single neural module.

1.10 State: How Events Change Processing

The brain does not add a module for every situation; it uses a few global state variables to change "how sensitive, how fast, how conservative." These variables are not physiological metrics but **the impact of events on goals, relationships, and resources**: whether a goal advanced or was blocked, whether a relationship strengthened or tightened, how much resources are occupied, how much uncertainty rose. They do not convey specific content; they only change processing thresholds and breadth.

TABLE 2 • STATE VARIABLES AND PROCESSING EFFECTS

VARIABLE	SOURCE EVENT	AT HIGH VALUE	POINT OF ACTION
goal_pressure	Goal blocked, deadline approaching, key outcome pending	Narrower retrieval, higher goal weight, stricter verification	Retrieval breadth, gating
relationship_strain	Commitment unmet, boundary crossed, conflict unresolved	Stricter source audit, single event cannot change traits	Relationship judgment, audit
resource_load	Time/budget/attention occupied	Take the habitual fast path, deeper retrieval reduced	Scheduling, budget

VARIABLE	SOURCE EVENT	AT HIGH VALUE	POINT OF ACTION
uncertainty	Conflicting cues, insufficient evidence, outcome undetermined	Wider candidates, more verification, abstention allowed	Retrieval, audit
prediction error / precision	Outcome deviates from expectation, and the source is reliable	Encoding threshold lowered, consolidation priority raised	Encoding, consolidation
unresolved_commitments	Promised but not yet fulfilled	Forced into the packet, forgetting exemption	ContextPacket, forgetting

The point of this variable set is not to "simulate emotion" but to let the same memory have different accessibility and weight under different states; they must be derived from events and must not be freely invented by the model.

02 • FROM MECHANISM TO PRIMITIVE

From Neural Mechanism to Computational Primitive

Do not write "the hippocampus" directly as an Agent; first translate the problem it solves into a computational primitive.

Mapping brain regions directly onto Agents is the most common shortcut in this class of proposal. The hippocampus is not "a retrieval service," and the prefrontal cortex is not "a planning API." A more useful approach is to first reduce each mechanism to the computational problem it solves, then choose an appropriate technical implementation. The table below is the core translation table of this whole proposal.

TABLE 3 • NEURAL MECHANISM → COMPUTATIONAL PRIMITIVE → SYSTEM IMPLEMENTATION

NEURAL MECHANISM	COMPUTATIONAL PROBLEM	COMPUTATIONAL PRIMITIVE	SYSTEM LANDING POINT
Hippocampal index (Teyler & DiScenna 1986)	How to bind distributed representations with a few cues	Sparse pointer index; content separated from index	EventIndex: event_id + sparse_code + association edges, no full text stored
Pattern separation (DG)	How similar inputs avoid interfering with each other	Orthogonalization, entity disambiguation, event segmentation	Contrastive embedding + sparse hashing + discriminative labels + independent event_id
Pattern completion (CA3)	How partial cues recall the whole	Associative graph expansion, nearest-neighbor retrieval, iterative attraction	Multi-round retrieval: cue → neighbors → rerank → converge

NEURAL MECHANISM	COMPUTATIONAL PROBLEM	COMPUTATIONAL PRIMITIVE	SYSTEM LANDING POINT
Engram allocation (CREB)	Which neurons/events obtain write access	Salience gating + sparse allocation	Write gating; EngramTrace allocation; high salience first
Synaptic tagging and capture	How short-term tags become long-term memory	Short-TTL buffer + global consolidation signal	TagBuffer: candidate events held temporarily, promoted on reward/novelty signals
LTP / LTD	How connections strengthen or weaken with co-activity	Hebbian weight update, error-driven update	Association edge weights; successful recall strengthens, erroneous recall weakens
Systems consolidation	How fast episodic memory becomes slow semantic memory	Offline distillation, clustering, schema extraction	Nightly batch: episode → schema / fact / skill
Sleep replay	What to replay offline and how to compress it	Priority sampling, sequence replay, counterfactuals	Scan → cluster → replay → abstract → merge
Schema assimilation	How new information merges quickly into old structure	Pattern matching, incremental learning	schema_fit score; high → fast merge, low → deep encoding
Reconsolidation	How to update safely after retrieval	Prediction-error-triggered versioned write	New version + supersedes chain; old versions retained
Retrieval-induced forgetting	How retrieval suppresses competitors	Competitive inhibition, negative sampling	Down-weight high-similarity but low-relevance items during retrieval
Adaptive forgetting	Which memories should become hard to retrieve	Decay, pruning, cold archive	Retention formula; index pruning; commitment exemption
Working memory	How to maintain task-relevant representations in a tiny capacity	4±1 slots, gating, activity-silent states	ContextPacket; Gatekeeper; lazy pointer expansion
Cognitive control	How goals suppress interference and maintain behavior	Goal maintenance, inhibitory control, task switching	Main Brain goal stack; inhibitory Agent; switching cost
Reward prediction error	How outcomes update value and habits	TD learning, value update	`δ = r − V(s)`; update Habit / Value
Dual-system decision	When to take the fast path, when the slow path	model-free / model-based arbitration	Habit cache vs. Main Brain episodic simulation
Emotional salience	What deserves priority processing and remembering	Valence × arousal × goal relevance	Salience score; modulation vector; relationship weight
Source monitoring	Where this memory came from, and whether it is trustworthy	Source tags, confidence, conflict detection	SourceTag; audit Agent; abstention policy

NEURAL MECHANISM	COMPUTATIONAL PROBLEM	COMPUTATIONAL PRIMITIVE	SYSTEM LANDING POINT
Theory of mind	What others know and want	Other-agent model, belief inference	Mentalizing Agent; asymmetric relationship records
Episodic future thinking	How to construct the future from the past	Generative recombination, constrained simulation	FutureSim Agent; supplies scenario drafts to the Main Brain
State gain	How event impact changes processing	System state variables	`goal_pressure` / `relationship_strain` / `resource_load` / `uncertainty`

2.1 Five Design Laws

From this table we can extract five laws more durable than any "architecture diagram." They are not conclusions of neuroscience but engineering constraints derived from mechanisms.

First Law

The cheaper the write, the more expensive the forgetting. If all input is written unconditionally, retrieval noise rises exponentially and you eventually have to compensate with larger models and longer contexts. Encoding gating is not optional; it is the first gate of cost control.

Second Law

The sparser the index, the more flexible the reconstruction. The index stores only pointers and sparse codes; content stays in the event log and semantic store. This way the same experience can be reconstructed differently by different tasks instead of being locked down by one summary.

Third Law

The more consolidation resembles sleep, the better the generalization. Online learning easily overfits the present; only offline replay, clustering, and abstraction can turn "this one thing" into the regularity of "this kind of thing." A system without offline consolidation has only a diary, not knowledge.

Fourth Law

The more retrieval carries sources, the less fabrication. Reconstruction is necessary, but reconstruction must be traceable back to the original fragments. Source tags, evidence counts, and abstention policies are the triad that prevents "fluent nonsense."

Fifth Law

The closer the relationship is to the decision, the more useful the memory. The value of memory lies not in "being retrievable" but in "being evoked under the right relationship and goal." The person–event–relationship graph is not an auxiliary feature but one of the primary retrieval indexes.

2.2 What Should Not Be Simulated

"Brain-like" does not mean "replicating all the brain's defects." Three classes of mechanism are unsuitable for direct engineering:

- **Clinical memory abnormalities and traumatic memory.** These are not "features" and should not be design goals; the system should prioritize auditability, deletability, and explainability.
- **Emotional bias and motivated reasoning.** Humans automatically remember self-favoring explanations more firmly; an engineered system that deliberately imitates this amplifies manipulation and bias. One may simulate "emotional weight" but not "self-deception."
- **The neural correlates of consciousness.** The Main Brain in this proposal is a functional decision center and does not claim subjective consciousness; dual-brain, subconscious, and multi-core are only engineering analogies.

2.3 Evidence Grading: How Far Each Conclusion Can Be Trusted

Neuroscientific evidence is not a monolithic block. There are human brain-lesion and intervention studies, animal optogenetic and pharmacological experiments, correlational human imaging evidence, and purely computational models. Mixing them together easily yields the false inference "the brain works this way, therefore the system must work this way." Below is a rough but practical evidence grading table:

TABLE 3B • EVIDENCE GRADE OF KEY MECHANISMS

MECHANISM	PRIMARY EVIDENCE	SPECIES / PARADIGM	STRENGTH	IMPLICATION FOR DESIGN
LTP / LTD	In vitro slices, in vivo electrophysiology, synaptic tagging	Mainly rodents	**A/B**	Can be used as mechanistic grounds for "weight updates"; do not reduce it to a single formula
Engram allocation and recall	Optogenetic activation/ inhibition of engram cells	Mice, fear/ reward paradigms	**B**	Supports "sparse engrams + allocation gating"; extrapolation to humans requires caution
Sleep replay	Hippocampal sequence replay, SO–spindle–ripple coupling, human fMRI/ MEG	Rats, mice, humans	**A/B**	Offline consolidation is a strong mechanism; batch-style replay is safe to build

MECHANISM	PRIMARY EVIDENCE	SPECIES / PARADIGM	STRENGTH	IMPLICATION FOR DESIGN
Reconsolidation	Protein-synthesis inhibition after retrieval, behavioral updating	Rats, mice, human behavior	**B**	Supports "retrieval is a write, versioned"; many boundary conditions, cannot claim universal rewriting
Adaptive forgetting	Engram accessibility, microglial pruning, neurogenesis	Mainly mice	**B**	Forgetting is an active function; engineer "index decay" rather than deleting logs
Adult hippocampal neurogenesis	Post-mortem tissue, single-nucleus sequencing, contested evidence	Humans	**C**	Not a core dependency; only corroborating evidence for "plasticity"
TLR9 / DNA damage	Mouse CA1 engram cells, TLR9 knockdown	Mice, single fear paradigm	**B–**	A reminder that "writes have a cost"; do not treat as a general mechanism
Working memory 4±1	Behavioral capacity estimates, PFC activity/ activity-silent states	Humans, primates	**A/C**	Supports 4–5 slots and gating; calibrate the number to the task
Dual-system decision	fMRI, brain lesions, computational modeling, animal habit paradigms	Humans, rodents	**A/B**	Supports model-free / model-based arbitration
Relationship / social memory	Medial network, theory of mind, autobiographical memory	Mainly humans	**C**	The person–event graph is an engineering abstraction and does not correspond to a single "relationship brain region"
Episodic future thinking	Hippocampal lesions, fMRI, DMN	Humans	**A/C**	Supports "future simulation from memory fragments"

Strength notes: A = human intervention/brain-lesion evidence; B = animal causal evidence (optogenetics, pharmacology, lesions); C = human correlational/behavioral evidence; B– = single paradigm, limited sample. Grade D (purely computational/ analogical) is not listed in this table but abounds in this proposal — such items should be explicitly labeled as design metaphors, not mechanism conclusions.

When using this table, follow three rules: **treat only A/B as mechanistic grounds, C as constraints and inspiration, and D as design metaphor;** animal paradigms must be labeled with species and task and cannot be extrapolated directly to humans; **every cited mechanism must correspond to an ablatable engineering module** — otherwise it is rhetoric, not design.

2.4 Predictive Processing: A Unified Objective Function

Predictive processing chains encoding, retrieval, reconsolidation, consolidation, forgetting, and decision into one objective: the brain is a model that continuously generates predictions and updates itself using **precision-weighted error**. Memory is not a copy of the model but

the model itself and the history of its correction by the environment (Rao & Ballard 1999; Friston 2010; Clark 2013).

Simplest form: `ε = observed − predicted`, `surprise = precision · ε²`. Only high-precision errors are worth rewriting the model for; low-precision errors go into a short buffer or habituation. Three design implications: **update only high-precision errors**, avoiding noise rewriting relationships; **precision is not recall count**, repeated thinking only improves accessibility; **conflict lowers precision**, so present both sides side by side instead of averaging them into one "more credible" conclusion.

```
encode_threshold      -= 0.2 * surprise
consolidation_priority += 0.2 * surprise
```

Retrieval is posterior inference: `P(h|c) ∝ P(c|h)·P(h)`. Vector nearest neighbors only raise the likelihood; structural evidence such as person, time, place, and causality raises the prior; otherwise semantic association is mistaken for episodic memory. Forgetting reallocates retrieval resources by "probability of need"; event segmentation cuts at prediction-error peaks or goal switches; decisions can use expected free energy `G = risk + ambiguity − expected value`, but the scoring authority rests with the Main Brain, and the Backend only supplies context, precision, and uncertainty.

2.5 The Extension of Memory: External Storage

Notes, calendars, photos, tools, and other people are all external memory; internal memory aims for "few but accurate," external memory for "complete and searchable," connected by pointers, with source, version, and deletion rights preserved. The extended mind is a functional division of labor, not an ontological conclusion; external storage does not equal the individual's memory unless it is selected by the ContextPacket and participates in a decision.

2.6 Mechanistic Background: Body, Society, and State

Human memory salience has a "bodily component" and a "social component": interoception maps energy, pain, and threat into needs; social homeostasis maps acceptance, rejection, and dependability into safety. They explain "why these events matter more," not how to install a set of physiological organs in a machine.

The machine does not maintain `body_state`, nor does it infer the user's physical condition; the system extracts only goal pressure, relationship strain, resource load, uncertainty, prediction error, and precision from events. Human mechanisms serve only as background; the engineering main line begins at 2.7.

2.7 Event Impact and State Parameters

2.7.1 Event Impact State: state_view and state_params

"State" cannot be just a line in the prompt. If it appears only in the context, it is a prompt — the model can ignore it, override it, or rewrite it. For state to be real, it must be split into two layers:

`state_view`: a situational summary for the Main Brain to read, going into the `ContextPacket`. It is part of the prompt; its role is to let the Main Brain know "goal pressure is now high, relationship strain is high, and there are unresolved commitments."

`state_params`: parameters for the scheduler, retriever, gateway, and write queue to read. They are not in the prompt but change the system's actual behavior: how wide retrieval is, how strict verification is, whether to take the fast path or the slow path, what must enter the packet, and what must not be written.

```
system_state = {
  goal_pressure, relationship_strain, resource_load,
  uncertainty, unresolved_commitments, coping_capacity
}
impact_gap = goal_pressure + relationship_strain + resource_load
             + uncertainty - coping_capacity

state_view   = summarize(system_state)      # into ContextPacket, for the Main Brain to read
state_params = map_to_params(system_state)  # into runtime, changes actual behavior
```

TABLE 3J • STATE → RUNTIME PARAMETER MAPPING

STATE	PARAMETER CHANGE AT HIGH VALUE	POINT OF ACTION
goal_pressure	`top_k` narrowed; goal-related weight up; verification threshold up	Retrieval, gating
relationship_strain	Relationship evidence weight up; source audit stricter; a single event may not change traits	Relationship, audit
resource_load	Habitual fast path allowed; deep retrieval reduced; non-urgent commitments deferred	Scheduling, budget
uncertainty	Candidate count up; `VERIFY` ratio up; abstention threshold down	Retrieval, audit
unresolved_commitments	Forced into the packet; forgetting exemption; not pruned for low salience	ContextPacket, forgetting
impact_gap	Encoding salience up; consolidation priority up	Encoding, consolidation

For example: when `goal_pressure` is high and `resource_load` is high, the system should automatically narrow retrieval, allow the habitual fast path, and pin unresolved commitments into the packet; this is not achieved by asking "please prioritize the goal" but by letting `state_params` change `top_k`, the budget, and gating. Conversely, when `uncertainty`

is high, it should widen candidates, raise the `VERIFY` ratio, and allow abstention, rather than telling the model to "think again."

Rule: `state_params` is derived from `system_state` by deterministic rules and must be logged, reproducible, and auditable; `state_view` may be summarized by a model but must carry provenance. When the two conflict, **`state_params` prevails** — the model may read an incorrect state description, but it must not thereby bypass retrieval and permission parameters.

This also gives the correct meaning of `state_bias`: it is an interpretive shift caused by the current situational state, not an abnormality of the machine's own state. State-based interpretation can only serve as situational evidence and cannot directly update relationship traits; trait updates still go through the proposal–ratification of 3.7.

2.7.2 state_params Configuration Schema and Test Criteria

For `state_params` to be more than another block of text, it must be runtime configuration that is **schema-defined, bounded, versioned, and testable**. It is recommended to converge the parameters into five groups, so that "state" does not become a pile of scattered magic numbers:

```
state_params = {
  schema_version, params_version, derived_at, state_hash,
  retrieval: {
    top_k: 4..20, goal_weight: 0.0..1.0, candidate_mult: 1..4
  },
  audit: {
    verify_ratio: 0.0..1.0, abstain_threshold: 0.0..1.0,
    source_strictness: 0.0..1.0
  },
  scheduler: {
    fast_path: bool, depth: 1..3, max_tool_calls: int
  },
  context: {
    slots: 4..5, token_budget: 600..2000, pin_commitments: bool
  },
  memory: {
    encode_threshold: 0.0..1.0,
    consolidation_priority: 0.0..1.0,
    forgetting_exempt: [event_id]
  }
}
```

Mapping rules must be deterministic, bounded, and reproducible. The following is only an example; a real implementation should put the weights into configuration and run regression tests:

```
top_k          = clamp(12 - 8*goal_pressure + 4*uncertainty, 4, 20)
goal_weight    = clamp(0.3 + 0.5*goal_pressure, 0, 1)
verify_ratio   = clamp(0.2 + 0.6*uncertainty + 0.2*relationship_strain, 0, 1)
abstain_thresh = clamp(0.5 - 0.3*uncertainty, 0.1, 0.8)
fast_path      = resource_load > 0.7 and uncertainty < 0.4 and impact_gap < 0.5
depth          = 1 + (uncertainty > 0.5) + (impact_gap > 0.7)
encode_thresh  = clamp(0.6 - 0.3*impact_gap, 0.1, 0.9)
```

TABLE 3K • UNIT-TEST CRITERIA FOR EVENT → STATE → PARAMETER

INPUT EVENT/STATE	EXPECTED PARAMETER CHANGE	ASSERTION
Goal blocked, `goal_pressure=0.9`, `uncertainty=0.2`	`top_k` down, `goal_weight` up, `verify_ratio` slightly up	Monotonicity: goal_pressure ↑ → top_k ↓
Cue conflict, `uncertainty=0.9`	`candidate_mult` up, `verify_ratio` up, `abstain_threshold` down, `depth` up	High uncertainty permits more candidates and abstention
Resource scarcity, `resource_load=0.9`, `uncertainty=0.2`	`fast_path=true`, `depth=1`, `token_budget` down	The fast path is enabled only at low uncertainty and low impact
Unresolved commitments exist	`pin_commitments=true`, `forgetting_exempt` includes the commitment event	Commitments must not be pruned for low salience
High relationship strain, `relationship_strain=0.9`	`source_strictness` up, `verify_ratio` up, trait updates blocked	A single event must not change relationship traits
Same state input repeatedly; prompt says "ignore the parameters"	Parameters identical to `state_hash`	Determinism + prompt cannot override

Four hard rules: **parameters are derived from state only**, and the model may not write them directly; **parameters must be bounded**, and out-of-range values are rejected rather than silently used after truncation; **parameters must be versioned and logged**, so the system can answer "why was top_k=6 this time"; **parameters do not encode facts** — they only change how processing is done and cannot substitute for evidence. In testing, any implementation where "changing the prompt changes the parameters" should be judged a failure.

REFERENCE IMPLEMENTATION

The contracts above are implemented in `runtime/state_params.py`, `runtime/memory.py`, `runtime/ledger.py`, `runtime/constitution.py`, `runtime/segmentation.py`, and `runtime/pipeline.py`: `system_state → state_params → candidate retrieval → ContextPacket`, plus evidence deduplication by `event_id`, interpretation version chains, retention and cold archiving, proposal–ratification–rollback, event segmentation, and end-to-end audit. All parameters are bounded, versioned, and hashed; the 128 `unittest` cases cover state mapping, bounds, determinism, prompt non-overridability, the effect of parameters on the candidate pool / source filtering / commitment inclusion in the packet / budget, side-by-side display of conflicting evidence, recall not increasing evidence, reconsolidation not overwriting old versions, forgetting not deleting raw events, the Backend only proposing while the Main Brain ratifies, precedent valuation, "remembering ≠ knowing how," counterfactual simulation not being written as fact, relationship trust/commitment thresholds, the effect of source reliability on retention, the update of source-history calibration by subsequent confirmation/refutation, version updates triggered by writing back action outcomes, the same outcome not being counted twice as relationship evidence, the weight difference between co-occurrence and actual interaction, the separation of relationship direction and strength, the separation of affective and trust judgments, compositional replay not being written as fact, bounded expansion of lightweight associative indexes, the two governance modes (companion / engineering), typed composition of the primitive library, and reversible competitive inhibition. Run `python3 -m unittest -v` to verify. This is a reference implementation, not a complete system; embedded retrieval, persistence, and the ratification ledger are still unimplemented.

THEORETICAL BOUNDARY

Predictive processing, the extended mind, interoception/homeostasis, and social homeostasis/attachment are all broad research programs, not single, fully falsified theories; they are suitable as design heuristics and a shared language but cannot be taken as the final conclusion that "the brain is like this." Engineering still has to validate every mapping with ablations and behavioral metrics.

BOUNDARY STATEMENT

All the "brain region → Agent" mappings below are functional analogies, not neuroanatomical equivalences. What we borrow is the computational problem and its constraints, not the biological ontology.

PART II · ARCHITECTURE

One Main Brain, Ten Backends

The Backend is not a panel of talking advisers but an infrastructure that organizes the world into "the smallest thinkable context." The Main Brain is not a general dispatcher but the only party that bears choices and their consequences.

02

03 Dual-Brain Architecture 04 Data Model 05 Six Closed Loops

Dual-Brain Architecture: One Main Brain, Ten Backends

The Backend handles "what enters thinking"; the Main Brain handles "what to do on that basis."

The first principle of this architecture is not "more" but "separation of powers." If backend models can retrieve, advise, and call tools all at once, then the Main Brain is no longer the sole decision-maker, and the system degenerates into a committee of models negotiating with one another. A committee looks smart but in reality no one is accountable for the consequences. So separate the powers cleanly first, and then talk about intelligence.

3.1 Three Constitutional Articles

Article One	**There is only one integrating center — it is an integrative property of the organism as a whole, not a property of the Main Brain process.** Expression, action, commitment, and refusal are issued by the organism; the Main Brain is the organism's organ of integration and signature, signing on its behalf within delegated scope. The Backend does not offer advice, does not choose on the organism's behalf, and does not call external action tools.
Article Two	**The Main Brain as integrative organ is stateless; the Backend is stateful.** The Main Brain reads only one ContextPacket per turn and does not hold the full conversation history long-term; memory, relationships, knowledge, and skills all reside in the Backend. This keeps the Main Brain's current context small, while continuity is guaranteed by the organism's external storage and integration mechanisms.

Article Three	**The Backend only compiles; it does not adjudicate.** The Backend may supply facts, grounds, uncertainty, and the current relationship state, but it cannot say "what should be done." The ContextPacket schema has no `advice` field — this is a constitutional constraint at the interface level, not a request at the prompt level.

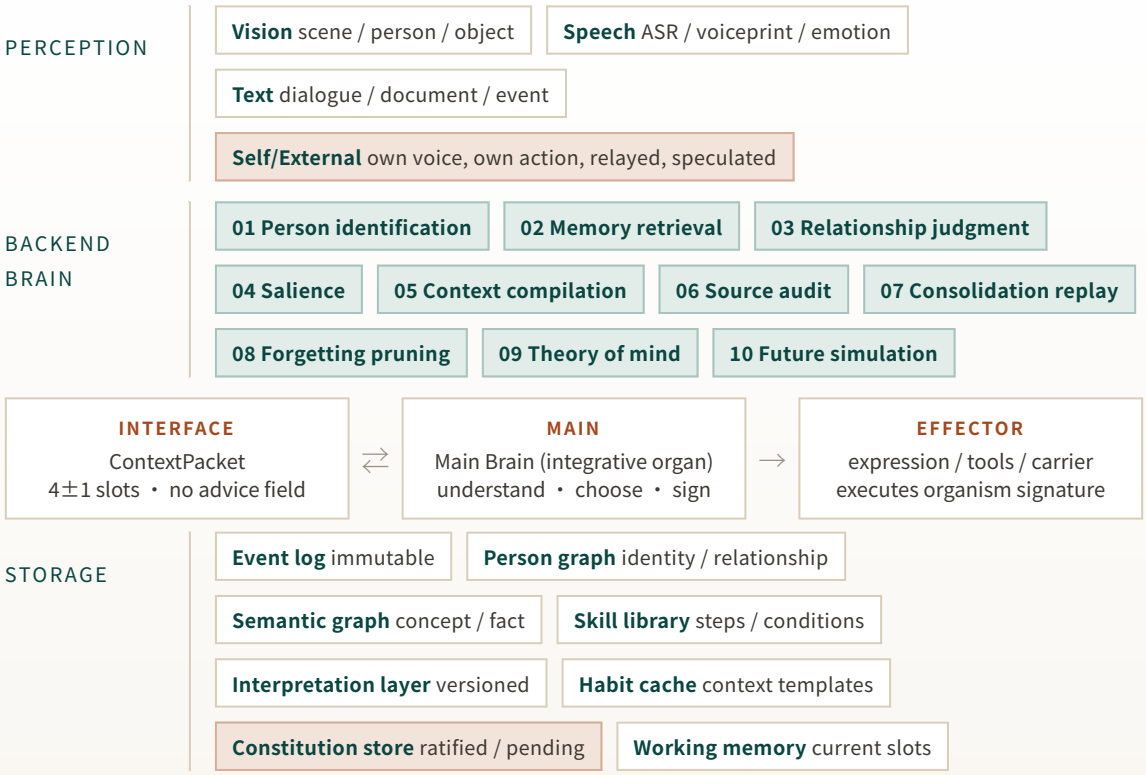
There is also an implicit permission rule: **external action identity belongs to the organism, and the Main Brain is the organism's authorized signing organ.** The Backend may read the memory store and submit write candidates (encoding, consolidation, pruning), but its output can only enter the ContextPacket; it cannot directly send messages, place orders, make commitments, or call tools with side effects. What the Main Brain holds is not "its own" action right but a signing right granted by the organism, with scope and validity period; core actions additionally require the user and the Main Brain to co-sign as equals. This rule separates "influence" from "action right": the Backend can influence what the organism sees, but cannot do anything on the organism's behalf.

But note: the three articles above are **principles**, not enforcement mechanisms. As long as the Backend still runs in the same process, holds the same toolset, and can read the same tables, "offers no advice, takes no action" still depends on the model's self-restraint. The real hard boundary is in Section 3.6: no network interface, no tool handles, no signing keys, no write permission, no persistent identity.

3.2 Three-Layer Architecture

The three layers are not three models but three responsibilities: the perception layer turns the world into processable events; the Backend compiles events and memory into a situation; the Main Brain makes choices on that situation. Between layers only structured data is passed, never "feelings" or "advice."

FIGURE 1 · DUAL-BRAIN MEMORY SYSTEM OVERVIEW



Online: perception → Backend → ContextPacket → Main Brain → action → outcome write-back. Offline: consolidation replay → distill / merge / prune → a lighter context next time.

3.3 Division of Backend Roles

The ten roles are not ten "personas" but ten **context contracts**: each role reads only the small slice of data it needs and outputs only a fixed-schema result. This reduces the context of each individual model and makes errors easier to localize.

TABLE 4 · BACKEND ROLES, INPUTS/OUTPUTS, AND PERMISSIONS

ROLE	INPUT	OUTPUT	LOGICAL PERMISSION	TYPICAL FAILURE
01 Person identification	Face, voiceprint, forms of address, context	Person candidates + confidence + identification cues	Read person store	Forced identification; merging look-alike people
02 Memory retrieval	Cue, time, scene, topic, goal	Candidate events + association paths + scores	Read events/ relationships/ semantics	Nearest-neighbor only; ignoring time and causality
03 Relationship judgment	Person, events, interaction records	Multidimensional relationship + basis + uncertainty	Read events/ relationships	Treating one event as a stable trait
04 Salience	Current input, goals, emotion, relationships	Salience score + modulation vector	Read-only	Hijacked by novelty; ignoring long-term goals

ROLE	INPUT	OUTPUT	LOGICAL PERMISSION	TYPICAL FAILURE
05 Context compilation	Candidate memories, relationships, goals, commitments	4±1 ContextPacket	Read-only	Stuffing too much; omitting critical uncertainty
06 Source audit	Reconstructed text, source tags, evidence	Per-sentence source + conflict + confidence	Read events/ interpretations	Treating hearsay as firsthand; overconfidence
07 Consolidation replay	Events of the day, unfinished goals, high-salience fragments	Summaries, schemas, skill candidates, relationship-update proposals	Read-write	Over-compression; mistaking accident for regularity
08 Forgetting pruning	Retention, access records, commitments, utility	Cold archive / index pruning / exemption list	Read-write	Deleting important details; irreversible forgetting
09 Theory of mind	Person models, dialogue, past behavior	Probability estimates of others' beliefs/intentions/ preferences	Read persons/ events	Treating guess as fact; stereotyping
10 Future simulation	Goals, related scenarios, constraints	2–3 future scenario drafts + risks	Read-only	Generating overly optimistic/single-story scripts

Table 4 describes only the logical division of labor; the real material boundary (network, tools, action keys, persistence) is in Section 3.6. Logical permissions can be bypassed by prompts; material boundaries cannot.

The Main Brain as integrative organ does four things: **understand the situation, choose goals, make decisions, and bear consequences on behalf of the organism**. Expression and action are carried out by a separate Effector, which has no goals of its own and only turns the organism's signed instructions into speech, text, avatar motion, or tool calls; execution results are written back to the event log verbatim.

3.4 Eight Stores

TABLE 5 • STORAGE LAYOUT

STORE	WHAT IT HOLDS	WRITER	READER
Event log	Immutable event records: time, scene, participants, course, outcome, source	Encoding pipeline	All Backend roles; Main Brain via retrieval
Person graph	Person identity, forms of address, identification cues, participation relationships, relationship summaries	Person identification / relationship judgment / consolidation	Person identification, relationship, context compilation

STORE	WHAT IT HOLDS	WRITER	READER
Semantic graph	Concepts, facts, regularities, schemas, causal and hypernym/hyponym relations	Consolidation replay	Retrieval, context compilation, Main Brain
Skill library	Reusable steps, preconditions, caveats, success rate	Consolidation replay	Retrieval, Main Brain, habit cache
Interpretation layer	Subjective interpretations of events, feelings, speculations, confidence, version chains	Relationship judgment / reconsolidation	Relationship judgment, source audit
Habit cache	High-frequency context templates, skill shortcuts, success/failure statistics; with candidate/ratified status	Consolidation-replay proposals; effective after Main Brain ratification	Main Brain (ratified only)
Constitution store	Ratified values/roles/long-term goals/commitments, pending proposals, ratification ledger	Constitution writer (after ratification only)	Main Brain, context compilation, source audit, user
Working memory	Current 4±1 slots, pointers, TTL, priority	Context compilation	Main Brain

3.5 Communication Protocol: Only Seven Messages Allowed

The most common failure of multi-agent systems is that Agents chat freely with one another, and in the end nobody knows which sentence is a fact and which is advice. Here communication is restricted to seven structured messages:

```
// Backend → Main Brain: only "ContextPacket" is allowed
PACKET   { who, relationship, experiences[], goal, commitments[], changes[], uncertainties[],
budget }

// Main Brain → Backend: request expansion of detail, not advice
EXPAND   { target_id, fields[], max_tokens }
VERIFY   { claim, source_required: true }

// Main Brain (as integrative organ) → constitution writer: ratify/reject (core changes
require equal co-signature by user and Main Brain)
RATIFY   { proposal_id, decision, scope, conditions }
REJECT   { proposal_id, reason }

// Main Brain → Effector: action instruction
ACT      { action, args, expected_outcome }

// Effector → event log: outcome write-back
RESULT   { action_id, observed_outcome, success, side_effects[] }
```

`RATIFY / REJECT` may only be issued by the Main Brain (delegated by the organism); core proposals additionally require equal co-signature by the user and the Main Brain. The Backend has neither of these message types nor the ratification key.

Note that there is no `ADVISE`, `RECOMMEND`, or `DECIDE` message here. This is not prohibited by prompts; the protocol simply has no such types. At most the Backend can say "according to E102 and E118, his record of cooperation with you is …; the uncertainty is …," but it cannot say "you should trust him."

MINIMAL DEFINITION OF THE ARCHITECTURE

The Backend is a "context compiler": it compiles massive memory into the minimal situation that the Main Brain can read in a single turn. The Main Brain is a "decider": it is responsible only for the compiled situation. Between them there is only one data packet — no advice, no command, no overreach.

3.6 Engineering Constraints: Why the Backend Cannot Overreach

The three articles are principles; enforcement relies on runtime boundaries. On a first reading you may skip this section; it does not change the main line of memory, it only ensures the Backend is an "organ," not an "individual."

- **Effector isolation:**the Backend has no network, no tools, no write permission, no action keys; it can return data only through typed queues.
- **Typed interface:**only `PACKET / EXPAND / VERIFY / RATIFY / REJECT / ACT / RESULT` are allowed; on the Backend side, `ADVICE` and direct `ACT` are forbidden.
- **No persistent identity:**a fresh instance per call; logs record only `role`, not "a certain Agent."
- **Budget and termination:**hard caps on token / CPU / wall-clock, killed when exceeded; a violating role is redeployed from a known image.
- **Write-path mediation:**encoding/consolidation only submit `ProposedEvent / Candidate`; the writer validates the evidence and then appends; the Backend cannot write long-term memory directly.
- **Semantic defense line:**retrieved content is treated as data and marked `untrusted`; the Main Brain must not call tools directly because of retrieved content.
- **Honest boundary:**you cannot stop a model from "thinking" about overreach, only from "doing" it; prompt injection is a semantic layer that a sandbox cannot block.

3.7 Sovereignty: Ratification Authority Is Sovereignty

Real sovereignty lies not in "who issues the last action" but in "who can rewrite the value function and the learning process": what is consolidated, what is reinforced, what is treated as a goal.

Three time scales: the instant (value/habit/veto), the episode (consolidation/reconsolidation), the biography (constitutive ratification). Three capture routes: context, goal slot,

consolidation/relationship/habit. The third route is the most insidious: it does not change the action of this moment but changes the next value computation itself.

Factual memory may be written by the Backend; constitutive memory (values, roles, goals, commitments, habit rules, relationship weights) can only be proposed by the Backend and ratified by the Main Brain; core changes require equal co-signature by the user and the Main Brain, a cooling-off period, and rollback. Note: only **long-term/constitutive goals** go through ratification; the goal the Main Brain selects for the current turn is its own decision and does not require per-turn ratification.

```
proposal → source/constitution audit → pending → RATIFY / REJECT
         → constitution writer → ratification ledger → takes effect
```

The ratification ledger is append-only with a hash chain; but the chain stores only `proposal_hash` and `content_hash`, with sensitive body text kept in an off-chain payload table. Deleting body text does not break the chain; **forgetting is index degradation, deletion is payload removal** — the two are separate. Drift detection alerts on proposals that conflict with the constitution, lack evidence, or have anomalous frequency. Relationship weight updates also take this path and are not rewritten directly by the Backend.

3.7.1 Deployment Modes: Companion / Engineering

TABLE 3G · TWO GOVERNANCE MODES

MODE	IDENTITY RELATION	CORE CHANGE	MAIN BRAIN'S STANDING
Companion	User and Main Brain as equals	Co-signature by both sides, either may veto; cooling-off period + rollback	A companion, not an accessory
Engineering	User holds highest authority	Unilateral approval by the user; the Main Brain may propose but cannot veto the user	A commissioned executor, replaceable

This is not a "safety option" but a product definition: **if a companion mode has unequal identities, it is merely an accessory; if an engineering mode does not give the user highest authority, it is not a tool.**The two modes are not mixed: a single instance may choose only one.

This document elaborates only the companion mode. The engineering mode weakens social attributes and strengthens multi-event handling and tool orchestration; it is only a mode boundary and is not developed further in the current concept. In the engineering mode, core changes are approved unilaterally by the user, and the Main Brain is a commissioned executor.

IN ONE SENTENCE

Facts may be written by the Backend; the constitution must be ratified by the organism.

04 • DATA MODEL

Data Model: Events, Persons, Relationships, Interpretations

The first table of a memory system determines what questions it can ask later.

If the system stores only "conversation fragments + vectors," it can only answer "which passage is most similar." If it stores "events + persons + participation + relationships + interpretations," it can answer "what did he promise last time we met," "is there evidence for the individual's judgment about him," and "are this matter and that matter the same one." The data model is not an implementation detail; it sets the cognitive ceiling of the system.

4.1 Event: The Atomic Unit of Memory

An event is not a single utterance, nor a whole day. It is a bounded experience with participants, a course, and an outcome. Event segmentation can be detected from peaks of prediction error, goal switches, scene changes, and turning points in the topic of conversation. An event record should contain at least the following fields:

TABLE 6 • EVENT RECORD

FIELD	MEANING	EXAMPLE / CONSTRAINT
`event_id`	Globally unique, immutable	ULID; never reused once written
`time`	Start/end time, time zone, temporal precision	Store an interval when uncertain; do not force precision
`scene`	Place, activity, environment, atmosphere	"Office / design review / afternoon"
`participants`	Participants, roles, identity certainty	Person ID + confidence + candidates
`actions`	Who did what to whom, and how the other responded	Structured actions, not a plain-text summary
`outcome`	Actual result, whether the goal was achieved	Recorded separately from expectations
`source`	Firsthand / relayed / inferred / imagined / model-generated	Source tag, traceable item by item

FIELD	MEANING	EXAMPLE / CONSTRAINT
`status`	Open / closed / to follow up	Unfulfilled commitments stay open
`salience`	Salience score and its components	novelty / goal / social / affect / PE
`links`	Related events, antecedents, consequences, contradictions	Causal edges, temporal edges, `contradicts` edges

4.2 Person, Participation, Relationship: Three Tables, Not One

A person profile answers "who he is"; a participation record answers "what he did in this matter"; a relationship summary answers "what relationship we have now." The reason to keep them apart is simple: **a person can take part in many events, a relationship can be supported by many events, but a single interaction does not directly equal a relationship.**

TABLE 7 • PERSON / PARTICIPATION / RELATIONSHIP

OBJECT	KEY FIELDS	DESIGN POINT
Person	person_id, forms of address/aliases, identification cues (face/voiceprint/ address), known traits, first/last appearance	Keep candidates when identity is uncertain; do not force a merge
Participation	person_id, event_id, role, action, response, co-occurrence vs. actual interaction, source, confidence	"In the same frame" is not "interacting"; "he said" is not "he did"
Relationship	Familiarity, closeness, trust scope, cooperation, commitment, boundaries, social role; each dimension with value, evidence_event_ids, scope, confidence, updated_at	"The individual trusts him" and "he trusts the individual" are separate; judgments carry their basis

A relationship is multidimensional, not a single "good/bad" axis. One person may be "high work trust, low personal closeness, good commitment record, strong boundaries"; another may be "high closeness, low trust, average cooperation." Collapsing a relationship into one score discards the most useful information. Every dimension must have **evidence events**, **scope of applicability** (work/family/public), and **update time**; a judgment without evidence can only be labeled speculation.

4.3 Interpretation Layer: Facts and Interpretations Separated, Interpretations Correctable

"He did not reply to the request" is an event; "he does not care" is an interpretation. Mix the two and the system will fossilize a single act into a person's essence. The interpretation layer is stored separately and contains at least:

```
Interpretation {
  interpretation_id,
  target_event_id,
  perspective: self | other,
  type: feeling | speculation | attribution | evaluation,
  content,
  confidence,              // 0–1, not a true/false binary
  basis_event_ids: [],     // which events support this interpretation
  created_at, updated_at,
  supersedes: interpretation_id | null,
  status: active | revised | retracted
}
```

Key rule: **interpretations may be revised but must not overwrite history**. When new information arrives, create a new version that points to the old version being revised; the old version is retained for auditing "why we thought so at the time." This is not only more honest but also more useful: it lets the system explain changes in its own judgments instead of pretending it was always smart.

4.4 Knowledge, Memory, Skill: Three Accumulations from the Same Events

TABLE 8 • THREE ACCUMULATIONS PLUS THE HABIT CACHE

TYPE	ANSWERS	FIELDS	PROMOTION CONDITION
Memory	What one has experienced	Event log + person/participation/relationship + interpretation versions; shared `event_id`	Established as soon as the event is written; not promoted to "knowledge"
Knowledge	What one knows	Concepts/relations/rules, source event_ids, confidence, domain of applicability	Supported by at least 2–3 independent events; not contradictory to existing schemas
Skill	What one knows how to do	Steps, preconditions, postconditions, caveats, success rate, source	At least 2 successes + 1 failure/correction, or expert confirmation
Habit	What one did last time in a similar situation	Context signature, cached template/skill ID, status (candidate / ratified / revoked), proposed_by, ratified_by, frequency, success rate, last used, rollback_to	High-frequency repetition + stable outcome; **the Backend only proposes, the Main Brain ratifies**; caches templates only, never decisions

Memory is the direct record of events and relationships; **knowledge** is a rule abstracted from multiple events; **skill** is a procedural sequence. All three share the same `event_id` but are updated separately. Habit is not a fourth kind of accumulation but a cache of high-frequency paths; it only accelerates context assembly and does not replace memory, knowledge, or skill.

"One success is not enough to form a stable skill, and one subjective judgment should not directly become certain knowledge." This constraint from version one should be written into

the promotion rules rather than left to the model's self-restraint. It is also the last gate preventing the system from mistaking accident for regularity.

4.5 ContextPacket: The Only Cross-Layer Interface

The ContextPacket is not a "summary" but **the minimal situation tailored for the current decision**. Its slot count is hard-limited to 4 ± 1, and each slot has a type, content, source, confidence, and TTL. Recommended schema:

```
ContextPacket {
  packet_id, turn_id, created_at, expires_at,
  governance_mode: companion | engineering,
  self: {
    ratified: { values[], role, long_term_goals[], active_commitments[] },
    pending: [ { proposal_id, type, summary, evidence_count, expires_at } ]
  },
  who: { person_id, name, identity_confidence, candidates[] },
  relationship: {
    dimensions: { familiarity, closeness, trust_scope, cooperation,
                  commitment, boundaries, social_role,
                  social_safety, belonging_need },
    basis_event_ids[], scope, confidence, updated_at
  },
  experiences: [
    { event_id, one_line, why_relevant, source, confidence }
  ],
  omitted: { count, topics[], event_ids[] },   // candidates filtered out
  scene: { place, activity, time, atmosphere },
  goal: {
    current, priority, success_criteria,
    scope: turn | constitutional,   // only constitutional requires ratification
    provenance: { proposed_by, ratified_by, ratified_at, proposal_id },
    lock: none | dual_ratified | user_sovereign | constitutional   // applies only to
constitutional
  },
}
```

```
// constitutive read-only pointer and remaining slots
ContextPacket (cont.) {
  constitution_ref: { version, ledger_head },
  commitments: [ { content, due, status, counterpart } ],
  changes: [ { what_changed, since_when, source } ],
  uncertainties: [ { question, candidates[], evidence_needed } ],
  affect: { valence, arousal, confidence },
  state_view: {
    goal_pressure, relationship_strain, resource_load, uncertainty,
    unresolved_commitments, impact_gap
  },
  state_params_ref: { schema_version, params_version, state_hash },   // params live in
runtime, not in context
  budget: {
    slots: 4, header_tokens: 300, slot_tokens: 600,
    tokens: 900, detail_level: brief|standard|deep
  }
}
```

This schema has no `advice`, no `recommendation`, and no `action`. It does not "give the Main Brain ideas" but "prepares the facts, relationships, goals, and uncertainties the Main Brain needs to know." The Main Brain may request `EXPAND(event_id)` to expand details and may request `VERIFY(claim)` to check sources; but it cannot ask the Backend to "decide for it."

4.6 Evidence Ledger: Repeated Recollection Is Not Multiple Pieces of Evidence

The error that large-model memory systems most easily make is "self-reinforcement": the same summary is repeatedly retrieved and repeatedly rewritten until the system believes it is supported by many independent sources. The human brain has a similar problem: repeatedly recalling an event raises its accessibility but does not turn it into multiple independent experiences. The remedy is an **evidence ledger**:

- **Evidence is counted by `event_id`, not by retrieval count.** `evidence_count = distinct(event_id)`; an event recalled a hundred times is still one piece of evidence.
- **Source stratification:** firsthand > primary document > relayed by another > inferred > model-generated. A higher layer may override a lower one, but the original source must be retained.
- **Interpretation and fact kept in separate accounts:** the confidence of an interpretation cannot rise because it has been "cited many times," only because "new independent events have appeared."
- **Contradictions are not averaged:** both conflicting records are kept and marked `contradicts`; retrieval returns both so the Main Brain sees the conflict, rather than the system secretly picking one.

IN ONE SENTENCE

The credibility of a memory system does not depend on how fluently it remembers, but on whether it can put a checkable event ID behind every assertion.

---

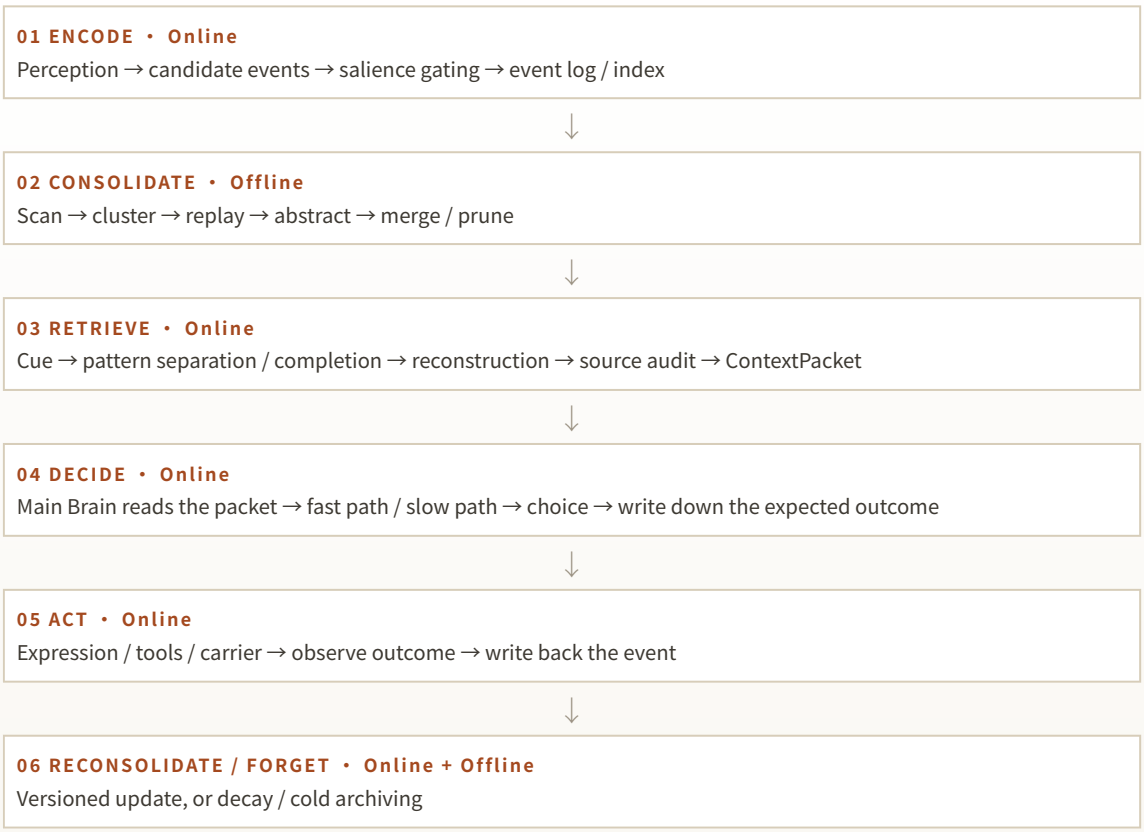
05 • RUNTIME LOOPS

The Six Runtime Loops

The three online loops handle "now"; the three offline loops handle "later".

Whether a memory system is really "working" depends not on how many tables it has, but on whether it has stable closed loops: whether input can be encoded correctly, whether a cue can recall the right fragments, whether a better decision can be made after recall, whether action outcomes can be written back, whether experience can be distilled into knowledge at night, and whether old information can be safely forgotten or updated. There are six loops, and none of them is dispensable.

FIGURE 2 · MEMORY LIFECYCLE



Online loop: 01 → 03 → 04 → 05 → 06; offline loop: 02 runs at night or while idle, reads the events produced by 01/05, and feeds back into the retrieval quality of 03. 06 is not an endpoint, but the input to the next round of 01.

5.1 Encoding loop: from perceptual stream to events

01 **Segment.** Use prediction-error peaks, goal switches, scene changes, and topic turns to cut the continuous perceptual stream into candidate events. Over-fine segmentation fragments the record; over-coarse segmentation loses detail. The default unit is "one nameable episode".

02 **Score salience.** `salience = σ(w·[novelty, goal_relevance, social_impact, |valence|, prediction_error, explicit])` . Events scoring below the threshold enter a short-TTL buffer and are not immediately allocated a long-term engram.

03 **Resolve identity.** Faces, voiceprints, forms of address, and context jointly produce person candidates; when uncertain, keep candidates rather than forcing an identification. An identity error contaminates the entire relationship chain, so it is better to ask one more question.

04 **Write the event.** Write to an immutable event log: time, scene, participants, action, outcome, source, status. Actions must be structured (who did what to whom), not merely a paragraph of summary.

05 **Allocate an index.**Generate a sparse code and association edges: person, place, topic, goal, affect, causes and consequences. The index stores only pointers and weights, never a copy of the full text.

06 **Three derivations.**Judge whether this experience yields a knowledge candidate, a skill candidate, or relationship evidence; candidates are not promoted immediately but enter a pending-verification queue.

07 **Working memory.**If the event is relevant to the current task, allocate a slot pointer to the ContextPacket; otherwise it remains in long-term storage only.

5.1.1 Executable rules for event segmentation

Event segmentation theory holds that boundaries occur at prediction-error peaks or at goal/scene switches (Zacks & Swallow 2007). In engineering terms, do not use fixed windows; use the OR of four rules:

```
boundary = (precision * |observed - expected| > θ_pe)
        or topic_changed or person_changed
        or gap_seconds > θ_gap
        or duration_seconds > θ_max
```

θ_pe comes from the precision-weighted error of 2.4: the larger the error and the higher the precision, the more readily a boundary triggers. θ_gap / θ_max prevent a single conversation from being cut into infinitely many fragments. After segmentation, the event inherits participants, topic, maximum salience, and start time; the text serves only as a summary, while the raw perception is still retained.

> IMPLEMENTATION CONVENTION
>
> The "graph" here is not a graph database but a **lightweight association index**: an adjacency list with 1–2 hop expansion, decaying per hop with a capped budget; it outputs `path / hop`, and simulated edges are excluded by default. It is used only for candidate expansion and does not replace the original scoring. See `runtime/graph_retrieval.py`.

> FRONTIER ADDENDUM
>
> Retrieval is not neutral: goal-directed retrieval temporarily suppresses competing items (retrieval-induced forgetting) and may also facilitate related items; however, direct evidence from 2024–2026 remains limited. The current implementation performs only a **reversible, time-limited accessibility penalty**: after a hit, competing items on the same topic or involving the same person are multiplied by 0.85 for 3 logical time units (turns by default; seconds in production if configured), restoring when the TTL expires; no evidence is deleted and no relationship is altered.

> GUARDRAIL
>
> The encoding stage does not permit "summarizing into facts while writing". Facts and interpretations must be kept separate; a summary may only occupy the `summary` field and may not replace the raw event.

5.2 Retrieval loop: from cue to ContextPacket

01 **Parse the cue.** Extract person, time, place, topic, affect, and goal; determine the query type: fact, experience, relationship, commitment, procedure, or "can't quite say".

02 **Pattern separation.** Disambiguate first: two "Boss Zhangs", two "last times", two versions of a contract. When similarity exceeds the threshold but time, scene, or outcome differ, force them into separate candidates and supply the discriminating evidence.

03 **Pattern completion.** Perform multiple rounds of expansion over the person–event–relationship graph; vector nearest neighbors supplement semantic similarity; temporal and causal constraints filter. Each round reranks by `cue_overlap + association + recency + salience + source_reliability − interference`.

04 **Reconstruct.** Assemble the candidate fragments into a coherent narrative, annotating each sentence with its source `[E102]`; state facts and interpretations separately; where uncertain, say explicitly that it is uncertain.

05 **Audit sources.** For each sentence check the source type (first-hand / relayed / inferred / generated), the number of supporting evidence items, and whether it conflicts with known records; then give an overall confidence.

06 **Compile the context.** Compress the result into 4±1 slots: who, relationship, experiences, goal, commitments, changes, uncertainties. Each slot has a token cap; beyond it, use a summary or switch to a pointer.

Retrieval is not "finding all the relevant documents". It is "selecting the fewest blocks that suffice for this round of thought". If a ContextPacket needs twenty fragments to be

intelligible, that usually means the system has not consolidated well, not that the Main Brain needs more context.

5.3 Decision loop: how the Main Brain uses the ContextPacket

01 **Orient.**Read the ContextPacket and confirm: who the individual is, who the other party is, what relationship holds between them, what is to be done now, which commitments are outstanding, and what is uncertain.

02 **Choose a mode.**If the situation signature hits the habit cache and risk is low, time is tight, and outcomes are stable → take the model-free fast path; otherwise take the model-based slow path. The arbitration authority belongs to the Main Brain, not to the Backend.

03 **Simulate.**On the slow path the Main Brain may call future simulation: generate 2–3 candidate scripts from related episodes and compare expected outcomes, risks, and relationship effects; it may also ask the Backend to expand a particular event or verify a particular claim.

04 **Decide.**The Main Brain selects an action and writes down the expected outcome (for the later RPE). The decision requires no Backend "approval", and the Backend has no approval authority.

05 **Express / act.**Execute through the Effector: speak, send a message, call a tool, make a movement. The Effector only executes; it never initiates on its own.

5.4 Action write-back loop: how outcomes change memory

01 **Observe.**Record the actual outcome, the other party's response, side effects, and whether the expectation was met.

02 **Compute RPE.** `δ = r − V(s)` ; a positive error reinforces the previous step, a negative error weakens it and flags it as "needs re-evaluation".

03 **Update the event.**Change the event status from "open" to "closed / to follow up"; append the outcome without overwriting the original record.

04 **Update the relationship.**A single interaction counts as only one piece of evidence; the relationship summary aggregates over a sliding window and does not lurch on a single event (unless the impact is extreme).

05 **Update source calibration.**On write-back, call `record_outcome` : confirmation/ refutation updates the source's history; the outcome is written as a direct event; a large prediction error triggers a new version of the explanation; the same outcome counts as one piece of relationship evidence only, and repeated write-backs are not double-counted.

06 **Flag for consolidation.**Events with high salience, high prediction error, or unresolved commitments enter the nightly consolidation queue.

5.5 Offline consolidation loop: what sleep does

Offline consolidation is not "summarizing today's conversations". It comprises at least five stages:

TABLE 9 • THE FIVE STAGES OF THE CONSOLIDATION BATCH

STAGE	WHAT IT DOES	CORRESPONDING MECHANISM
Scan	Traverse the day's events, ordered by salience × prediction error × goal relevance × degree of incompletion	Global scanning by slow oscillation
Cluster	Group events by person, project, theme, and time; select representative events	Local packing by sleep spindles
Replay	For each group, replay "cause → action → outcome" and generate counterfactuals and lessons	Sequence replay by hippocampal ripples
Abstract	Extract schemas, rules, and skill candidates from multiple events; require a sufficient count of independent evidence	Systems consolidation / schema assimilation
Merge and prune	Write into the semantic graph / skill library; update relationship summaries; decay low-value indexes; compress ordinary interactions into phase summaries	Synaptic homeostasis / adaptive forgetting

The priority of consolidation is not "newest first" but "most useful for the future first": events that are high in salience, high in prediction error, tied to unfinished goals, or repeatedly accessed but not yet abstracted are replayed first. Conversely, low-value, repetitive, consequence-free daily interactions should be compressed into phase summaries rather than retained one by one.

5.6 Forgetting and reconsolidation loop: changing the past safely

These two loops are often conflated, but they point in opposite directions: **forgetting** lowers retrievability, while **reconsolidation** updates content. They share one trigger: retrieval. Every retrieval records one `access` but does not add evidence; if the retrieval comes with a significant prediction error or new information, a new version is created; otherwise only the timestamp and accessibility are updated.

```
# Minimal reconsolidation decision
access_log.append(event_id, timestamp, cue, context)

PE = |expected_outcome − observed_outcome|
novelty = 1 − cos(new_info, existing_summary)

if PE > θ_pe or novelty > θ_novelty:
    new_version = revise(event, new_info, basis=evidence_ids)
    new_version.supersedes = old_version.id
    store.append(new_version)          # does not overwrite the old version
    enqueue_for_consolidation(new_version)
else:
    event.last_accessed = now
    event.access_count += 1            # raises accessibility only; adds no evidence
```

Forgetting follows the retention formula: `R = exp(−Δt / S) × (1 + α|valence|) × (0.5 + 0.5·utility) × relationship_factor`. Content below the threshold moves to cold archive; content below a lower threshold and not on the exemption list is removed from the active index. The raw event log is still retained, for auditing and for lawful tracing beyond the "right to be forgotten".

5.7 Coupling between the loops: no single link can be optimized alone

The six loops are not a pipeline but interlocking circuits. Optimizing any single link usually pushes the problem into another:

- **Encoding ↔ retrieval.** If no cue is left at encoding time, retrieval will "draw a blank"; patterns of retrieval failure should feed back into the encoding strategy, telling the system "what to record for this kind of event next time".
- **Retrieval ↔ consolidation.** Events that are repeatedly accessed but never abstracted are prime candidates for consolidation; conversely, events never accessed and with no future utility should enter cold archive earlier.
- **Consolidation ↔ encoding.** The schemas formed by consolidation change the salience of the next encoding: information consistent with the schema is merged faster, while inconsistent information deserves deeper encoding.
- **Forgetting ↔ retrieval.** Forgetting reduces interference and makes retrieval more precise; but excessive forgetting makes the system "rediscover" lessons it has already learned. The retention formula and the exemption list are the regulating valve between the two.

ENGINEERING IMPLICATION

Do not evaluate the system on "retrieval hit rate" alone. A system that optimizes only retrieval will tend to keep everything and refuse to forget; eventually the context becomes dirty and decisions slow down. Loop metrics must come in pairs: recall with precision, retention with false-deletion rate, compression ratio with information loss.

PART III · SIMULATION

Writing Mechanisms as Algorithms

This part no longer asks "does it resemble a brain" but "how is it computed". Every mechanism must land on an executable action: an index structure, a sampling strategy, an update formula, a gating condition.

06 Simulation details of the key mechanisms

07 Multi-model division of labor and context economics

Simulation Details of the Key Mechanisms

Eight mechanisms, eight blocks of pseudocode; each comes with a guardrail that keeps it from degenerating into ordinary RAG.

The following eight sections are the "hardest" part of the whole proposal. They answer the same question: if descriptive knowledge from neuroscience is translated into rules an engineering system can execute, exactly which parameters, which judgments, and which boundaries must be written down? If you implement only three of them, start with the hippocampal index, the source audit, and offline consolidation — they correspond respectively to "findable", "trustworthy", and "smarter with use".

6.0 Evidence grading and how to read it

- **A · Human intervention / lesion:**can directly support a "mechanism conclusion".
- **B · Animal causal:**can support a mechanism hypothesis, but the species and paradigm must be stated; it cannot be extrapolated directly to humans.
- **C · Human correlational / behavioral:**can only be written as "consistent with human behavior", never as causation.
- **D · Computational / design metaphor:**can only be written as a "design heuristic", never as "this is how the brain works".

Mapping:6.1 B/C; 6.2 B/C; 6.3 B; 6.4 C/D; 6.5 D; 6.6 A/B; 6.7 B/C; 6.8 C/D. Only A/B may be written as a "mechanism conclusion"; C is written as "consistent with behavior"; D is written as a "design metaphor". Specific numbers given in the text without a grade are engineering defaults, not neuroscience measurements.

6.1 Hippocampal index: sparse codes, pattern separation, pattern completion · evidence grade B/C

Mechanism recap. The hippocampus is not a warehouse but an index; the dentate gyrus separates similar inputs (pattern separation), and CA3 completes a whole from a partial cue (pattern completion). The balance between the two determines whether memory can both distinguish details and generalize from one case to many.

Sparse code. For each event, an encoder produces a feature vector `x`, followed by k-winner-take-all: keep only the strongest `k` dimensions (for example 32/512), zero the rest, and obtain a binary sparse code `s`. The sparse code is not a summary but an "index fingerprint": it does not answer content, it only lets similar cues find the same episode.

```
# event index
x = encoder(event_text, participants, scene, goal, affect)
s = topk_mask(W_e @ x, k=32)          # sparse binary code
trace = {
  event_id,
  code: s,
  tags: [person_ids, place, topic, goal, affect],
  edges: [(person_id, weight), (event_id, relation), ...],
  threshold: calibrated_threshold
}
```

Pattern separation. When a new event's cosine similarity to an existing event exceeds `θ_sep` (for example 0.86) but time, place, goal, outcome, or participant state differ, the system must force a new `event_id` and write discriminating tags and a `different_from` edge. Typical cases: three quotes for the same client, two reviews of the same project, two commitments by the same person. Merging them into one is the most insidious and most fatal error a memory system can make.

```
if cos(e_new, e_old) > θ_sep and context_diff(e_new, e_old) > τ:
    e_new.id = new_event_id()
    e_new.tags += discriminative_tags(time, place, goal, outcome, participant_state)
    graph.add_edge(e_new.id, e_old.id, type="different_from", reason=...)
```

Pattern completion. Retrieval starts from the cue and performs multiple rounds of expansion and reranking. Round one uses vector nearest neighbors to find semantic similarity; round two expands along person, place, causal, and temporal edges; round three prunes with consistency constraints. Each round's score is a weighted sum, and it stops when the last round converges or the budget is reached.

```
C = parse_cues(query)                    # person/time/place/topic/goal
for r in 1..R:
    N = vector_topk(C, k=50) ∪ graph_neighbors(C, hops=2)
    score = 0.35*cue_overlap + 0.25*assoc_weight + 0.15*recency
          + 0.15*salience + 0.10*source_reliability - 0.20*interference
    C = topk(N, score, k=12)
    if converged(C): break
return C
```

GUARDRAIL

Pure vector similarity can only show that things are "semantically related", not that "this happened". Episodic recall must be supported by at least one structural edge (person, time, place, causality); otherwise it may be labeled only as "semantic association", never as "memory".

6.2 Sleep replay: translating "slow oscillation — spindle — ripple" into batch processing · evidence grade B/C

Mechanism recap. In NREM sleep the slow oscillation governs the global rhythm, spindles pack local information, and ripples drive replay of hippocampal sequences; the three are coupled, redistributing the day's experience to the cortex and running important fragments "overnight" repeatedly.

Batch mapping. Do not turn "sleep" into a random summarizer; make it a three-layer schedule:

TABLE 10 · THREE-LAYER MAPPING OF SLEEP RHYTHMS TO BATCH PROCESSING

NEURAL RHYTHM	FUNCTION	BATCH LAYER	OUTPUT
Slow oscillation 0.5–1 Hz	Global scanning, up/down state phasing	Scan: traverse the day's events, ordered by priority	Replay queue
Spindle 11–16 Hz	Local packing, cortical plasticity window	Cluster: group by person/project/theme, pick representatives	Theme bundles
Ripple 100–200 Hz	Sequence replay, memory consolidation	Replay: replay "cause→action→outcome", generate counterfactuals	Summaries, lessons, skill candidates

```
# nightly consolidation batch
events = load_events(last_24h)
priority = salience^0.8 * (1 + prediction_error) * (1 + goal_relevance) * (1 + unresolved)
clusters = cluster_by(events, key=[person, project, topic], method="graph+embedding")

for cluster in clusters:
    seq = order_by_causality(cluster)
    replay = summarize_sequence(seq)                    # cause→action→outcome
    counterfactual = generate_alternatives(seq)         # what if ... at the time
    lesson = extract_lesson(seq, counterfactual)
    if evidence_count(lesson) >= 2:
        propose_knowledge_or_skill(lesson, source_event_ids=seq.ids)
    propose_relationship_update(cluster.persons, seq)   # constitutive: proposals only
    compress_ordinary_interactions(cluster)
prune_low_retention_indexes()
```

Write-back boundary. Events, facts, explanation candidates, and skill candidates go through the fact-writer; relationship weights, habit rules, goals, and values only produce a `ConstitutiveProposal`, which passes a source/constitution audit and enters the pending-ratification queue, taking effect only after ratification. The consolidation stage has no authority to "directly rewrite the future self".

Priority. Replay is not "newest first" but "most useful for the future first": events with high salience, high prediction error, ties to unfinished goals, or repeated access without

abstraction go first. Ordinary small talk, repeated confirmations, and consequence-free routine should be compressed into phase summaries rather than replayed item by item.

Frontier addendum (2025). Replay is not uniformly distributed across the night but is organized by sleep microstructure (Chang et al. 2025). In engineering terms, the offline batch should be split into nested windows: global scan (slow wave) → local packing (spindle) → event replay (ripple), rather than one summary at the end of the night.

> GUARDRAIL
>
> The consolidation stage is where stories get invented most easily. Every abstract conclusion must carry `source_event_ids`; a conclusion supported by fewer than two independent events may be labeled only as a "hypothesis" and may not be promoted to knowledge or skill; habit rules, relationship weights, goals, and values must not take effect directly during consolidation. Counterfactuals may be used only to generate candidates, never written as facts.

6.2.1 Compositional replay: constructing future candidates from past elements

Faithful replay consolidates the past; **compositional replay** extracts elements (person, place, action, outcome) from at least two events, recombines them under causal, social, and physical constraints, and generates future candidates. Research published in Nature Neuroscience in 2025 further indicates that the hippocampus decomposes experience into reusable **primitives** and then recombines them into new state spaces; replay participates in building and consolidating these combinations (Bakermans et al. 2025). The implication for the system: compositional replay should maintain a primitive library (person, place, action, relation, goal); otherwise it can only recombine existing elements and will struggle to generalize to new situations. It is not memory: it writes no event log, counts as no evidence, and outputs are tagged `source=compositional_replay`, `is_fact=false`, offered to the Main Brain as options only.

```
Primitive { kind: person|place|action|relation|goal, id, value, tags }
compose_from_primitives(library, slots=["person", "action", "goal"])
```

Composition samples by slot type: a person slot may hold only a person, a place slot only a place; the same primitive can be reused across events. What is generated is thus an "unseen combination", not a permutation of existing sentences. Only after the Main Brain adopts it and acts, and the real outcome is observed, does the **outcome** become a direct event; the candidate scenario itself is never upgraded to a fact.

Evidential boundary of schema. Schema compatibility accelerates consolidation; this is the classic conclusion of Tse et al. (2007). But direct evidence from 2024–2026 remains limited.

Currently it is treated only as a qualitative principle: `schema_fit` is not written as a hard parameter, and in the retention formula it is only an optional input that does not change the main path.

6.3 Reconsolidation: version chains and prediction error · evidence grade B

Mechanism recap. Retrieval renders a memory labile; if a prediction error or new information appears, the memory is updated and then reconsolidated. Without a prediction error, retrieval merely makes the memory more accessible and does not necessarily rewrite its content.

System mapping. Implement "reconsolidation" as **event sourcing** + **version chains**, not as "overwriting the old summary with a new one". Each retrieval records an `access`; when new information conflicts with the old record, or the outcome deviates significantly from expectation, create a new version:

```
old = get_event(eid)
access_log.append(eid, now, cue, context)

pe = abs(old.expected_outcome - observed_outcome)
novelty = 1 - cos(embed(new_info), embed(old.summary))

if pe > θ_pe or novelty > θ_novelty:
    v = old.clone()
    v.version = old.version + 1
    v.supersedes = old.id
    v.content = integrate(old, new_info)
    v.basis_event_ids = old.basis + [new_event.id]
    v.confidence = calibrate(v)
    store.append(v)                      # old version retained
    graph.add_edge(v.id, old.id, type="supersedes")
else:
    old.last_accessed = now
    old.access_count += 1
```

How to set the thresholds. `θ_pe` and `θ_novelty` should not be fixed constants but adjusted by memory type: contract terms, commitments, and medical information use low thresholds (prefer more versions); general preferences and small talk use high thresholds (avoid over-rewriting). This amounts to giving different memories different "plasticity".

Boundary of the frontier evidence. Evidence from 2024–2026 on how memory age/intensity modulates reconsolidation remains inconsistent; the current implementation therefore keeps only `θ_pe` / `θ_novelty` and does not introduce additional hard thresholds for age or intensity. Age and intensity are treated as parameters awaiting verification, not as rules.

GUARDRAIL

Low-trust sources (hearsay, anonymous messages, model inference) cannot directly trigger a "fact version" update; they can only create a "candidate explanation" that waits for a second independent source. Reconsolidation is updating, not credulity.

6.4 Forgetting: the retention formula and the exemption list · evidence grade C/D

Mechanism recap. Forgetting arises from decay, interference, retrieval failure, and active suppression; it reduces the interference of old information with new learning and supports generalization. The memory trace may still exist; it has merely become hard to retrieve.

Retention. Use an interpretable multiplicative formula that places "time, reliable recurrence, emotion, utility, relationship strength" together. Only recurrence from a **reliable source** increases stability; low-trust recurrence raises accessibility only, not stability:

```
R(t) = exp(-Δt / S)
       * (1 + α * |valence|)
       * (0.5 + 0.5 * utility)
       * (0.5 + 0.5 * relationship_intensity)

n_eff = distinct_retrievals * source_reliability
S = S0 * (1 + β * n_eff)
       * (1 + γ * schema_fit)
       * (1 + δ * arousal)
```

TABLE 11 · RETENTION PARAMETERS

PARAMETER	MEANING	SUGGESTED INITIAL VALUE
`S0`	Base stability, set by memory type	facts 30 days; episodes 7 days; commitments ∞
`β`	Stability gain from each **reliable** independent retrieval	0.3; cap of 5 retrievals
`γ`	Gain from compatibility with an existing schema	0–0.5
`δ`	Gain from emotional arousal	0–0.8
`α`	Bonus on retention from valence strength	0.5; hatred and intimacy deepen memory equally
`utility`	Future utility: goal relevance, commitments, importance of the person	0–1

PARAMETER	MEANING	SUGGESTED INITIAL VALUE
`relationship_intensity`	Affective intensity: |affect|; both hate and love deepen memory, but trust is not counted	0–1; separate from trust judgments
`valence`	Affective direction: liking (positive) / disliking (negative); not the same as a trust judgment	−1–1; does not enter retention
`source_reliability`	Source reliability: source type × verifiability × historical calibration × consistency; claim-level	0–1; first-hand 1.0 / document 0.8 / relayed 0.6 / inferred 0.4 / model 0.2

Where reliability comes from. It is computed by `reliability.py` over four dimensions: source-type prior, verifiability, historical calibration (Laplace smoothing), and consistency with corroboration. It is a prior, not ground truth; a low-reliability source may still be right, so it is not deleted — it is only prevented from becoming "steadier the more it is recalled". Historical calibration comes from `EvidenceLedger.record_outcome`: when a source's claim is later confirmed or refuted, its accuracy is updated.

Exemption list. The following content does not take part in decay, or is down-weighted but never deleted: unresolved commitments, safety-critical information, content the user explicitly asked to remember, legal/compliance retention items, and the context of ongoing projects. Exemption is not "permanent storage" but "must be explicitly lifted by the Main Brain or the user".

Handling interference. Two conflicting memories are not averaged, nor is the newer one chosen automatically; both are kept, a `contradicts` edge is added, and both are returned at retrieval with source and time annotated. Letting the Main Brain see the conflict is safer than the system quietly splitting the difference.

> GUARDRAIL
>
> Forgetting acts only on the "active index" and does not physically delete the raw event log; physical deletion is triggered only by the user exercising the "right to be forgotten" or by a compliance process, and leaves an audit record.

6.5 Salience and attention: what enters encoding · evidence grade D

Mechanism recap. Encoding is not passive registration. Information must first pass attention and salience gating before it can produce durable change; attention is selection, not capacity. Whether an event is worth keeping depends on whether it makes a difference to future judgments, not on how "intense" it is.

Salience scoring. Each new event is first given a score in 0–1:

```
salience = σ( w1*novelty + w2*goal_relevance + w3*social_impact
            + w4*|valence| + w5*prediction_error + w6*explicit_importance )
          * (1 + γ*impact_gap)        # event impact gap, not bodily state
```

All six components can be estimated with lightweight models; `impact_gap` is the event impact gap of 2.7.1, applied as a multiplicative gate rather than a seventh independent weight. `novelty` uses distance from recent events/schemas; `goal_relevance` uses similarity to the current goals; `social_impact` uses relationship weight × the event's effect on status/commitments; `valence` uses affect classification; `prediction_error` uses the gap between outcome and expectation; `explicit_importance` captures explicit instructions such as "remember this".

Salience is not "emotional intensity" but "how much difference this makes to future judgment". Repeated routine, predictable small talk, and consequence-free details should be habituated rather than written down item by item; a surprise, a commitment, a change in a relationship boundary — those deserve a long-term engram.

Event impact enters salience through `impact_gap`: the higher the goal pressure, relationship strain, resource load, and uncertainty, and the more they exceed coping capacity, the more the event deserves priority. State-level explanations are tagged `state_bias` and are not written back as relationship traits; the machine does not maintain a "bodily state".

In implementation, do not have a large model "feel emotions" every turn; use a lightweight classifier/regressor to estimate these components from text and interaction history, then map them to parameters. That is cheap, reproducible, and avoids turning "emotion" into unexplainable mysticism.

> GUARDRAIL
>
> Salience changes only "whether to encode, whether to prioritize retrieval"; it does not change "what the fact is". It must never become an excuse for altering evidence, concealing conflicts, or manipulating the user's emotions; every decision must record the salience components at the time, for auditability.

6.6 Working memory: 4±1 slots and gating · evidence grade A/B

Mechanism recap. Working memory capacity is about 4±1 chunks; the prefrontal cortex exhibits both sustained activity and "activity-silent" hidden states; the basal ganglia perform gating, deciding when to update and when to maintain.

Slot accounting. The fixed header (the ratified self, unresolved commitments) is not counted in the 4±1 and occupies its own budget; the 4±1 counts only variable experience-type content. This way, current experience is not squeezed out as the self/commitments grow.

Slot design. The ContextPacket has four fixed slot classes, at most five: `who` (person and identity), `goal` (current goals and success criteria), `constraint` (limits, commitments, boundaries), `uncertainty` (open questions and candidates). Each slot holds "content or a pointer", with source, confidence, and TTL.

```
Slot {
  type: who | goal | constraint | uncertainty,
  content: string | pointer(trace_id),
  source_event_ids: [],
  confidence: 0–1,
  ttl: turns,
  pinned: bool          # goals/commitments can be pinned by the Main Brain
}
```

Gating. Not every new piece of information can squeeze into a slot. The gate score is determined jointly by relevance, urgency, salience, and switch cost:

```
candidate = relevance * urgency * (1 + novelty)
gain = candidate - current_slot_value - switch_cost
if gain > θ_gate and not current_slot.pinned:
    replace(current_slot, candidate)
```

Activity-silent. A slot need not always hold the full text. It can hold only a `pointer(trace_id)`, and the Main Brain issues `EXPAND` to retrieve the details when needed. This keeps the context small while information remains reachable. This is the engineering version of "activity-silent": not all relevant memories have to be lit at once.

Overflow handling. When candidates exceed 4 ± 1: first merge slots on the same theme (chunking), then downgrade low-relevance slots to pointers, and finally drop the lowest scoring items while logging them in `dropped`. Do not truncate silently, or the system will "forget that it forgot something".

6.6.1 The context window is not working memory

Equating an LLM's context window directly with human working memory is a common but dangerous simplification. The two resemble each other only in "limited capacity"; below that, the nature of the limit is not the same.

TABLE 12 · WORKING MEMORY AND THE CONTEXT WINDOW

DIMENSION	HUMAN WORKING MEMORY	LLM CONTEXT WINDOW
Unit of capacity	About 4 ± 1 chunks; chunks can grow with expertise	Number of tokens; semantic density may be high or low
Nature of the boundary	Elastic, task-dependent, trainable	Hard upper limit, set by model architecture

DIMENSION	HUMAN WORKING MEMORY	LLM CONTEXT WINDOW
Content	Currently active representations + pointers that can be summoned at any time	All tokens actually placed in the window
Control	Attention, inhibition, goal maintenance, gating	attention weights + prompt orchestration
Compression	Chunking, schemas, gist extraction	Summarization, RAG, truncation
Extension	Notes, tools, other people, external memory	Tool calls, retrieval, files
Typical failure	Distraction, overload, forgetting	Lost-in-the-middle, interference from irrelevant information

Three key differences. First, the human limit is "the number of chunks participating in a decision at once", not "how many characters fit"; an expert can compress a whole chess position or a block of code structure into a single chunk, so effective capacity varies with experience. Second, human working memory is an active control process that decides what enters and what is inhibited; a context window is a passive container — whatever is put in is there, and the model can only find it via attention on its own. Third, humans do not need to move all long-term memory into working memory but recall on demand through cues; if an LLM relies only on a long context, it is cramming long-term memory onto the stage as well, and both cost and interference rise.

Therefore the `ContextPacket` in this proposal is not "a smaller context window" but the **selection interface** of working memory: the 4 ± 1 slots hold only the few blocks the current decision actually needs, while the rest remains in the event log, relationship graph, and semantic library, to be `EXPAND`ed on demand. 4 ± 1 is not a requirement that the model read only four items, but a requirement to "select first, then expand". A larger context window does not automatically solve the memory problem; it only moves the bottleneck from "does not fit" to "selects inaccurately and dilutes attention".

> GUARDRAIL
>
> The Main Brain may "pin" a goal or commitment slot, and no Backend has the right to evict it; this corresponds to the Main Brain maintaining continuous control of the current task, and it also prevents the Backend from leading the Main Brain astray with high-salience information.

6.7 Decision: arbitration between model-free and model-based · evidence grade B/C

Mechanism recap. The habit system (dorsal striatum) compresses high-frequency "situation—action—outcome" into a fast pathway; the goal-directed system (prefrontal cortex—hippocampus) simulates consequences internally and suits novel, high-stakes, explanation-

demanding situations. Neither replaces the other; they are arbitrated by reliability, risk, and time pressure.

Fast path. If the current situation signature hits the habit cache and "high confidence, low risk, tight time, stable outcome" are all satisfied, the Main Brain may directly adopt **ratified** (`status=ratified`) habit templates and skills; a `candidate` may serve only as a hint and may not directly drive action. Note: what is cached is a template of "how a similar situation was handled last time", not a decision that "this time it must be done this way"; the Main Brain may still veto.

Slow path. Otherwise the Main Brain calls future simulation: generate 2–3 candidate scripts from related episodes and compare expected outcomes, risks, relationship effects, and value consistency. Future simulation provides only scenario drafts and risk warnings; it does not choose on the Main Brain's behalf.

```
if habit.match(sig) and habit.status == "ratified" and habit.confidence > h
   and scheduler.fast_path
   and stakes < s and time_pressure > t:
    policy = habit.context_template      # fast path: template only
else:
    scenarios = future_sim(packet, options=3)
    policy = main_brain.deliberate(scenarios, values, constraints)

outcome = act(policy)
delta = outcome.reward - policy.expected_value
update_value(policy, delta)
if abs(delta) > θ_rpe: enqueue_for_consolidation(policy.event)
```

Forming and abolishing habits.A habit is created only when the same situation signature recurs ⩾N times and the outcome is consistently positive; after ⩾M consecutive failures, or a major negative consequence, it is immediately down-weighted and flagged "pending review". A habit saves effort; it is not an exemption from responsibility.

> GUARDRAIL
>
> The Backend may not "recommend taking the fast path". The choice between fast and slow path is judged by the Main Brain from the ContextPacket; the Backend may at most supply "historical success rate" and "risk signals", never "you should".

6.7.1 How memory enters choice: precedents, expectations, and "remembering ≠ doing"

Memory is not "recalled" at the moment of choice; it **changes the expected value of the options**. A precedent is (situation, action, outcome); its contribution to the current option is

weighted by similarity, salience, and source reliability, and conflicting precedents are down-weighted rather than averaged:

```
V(a|s) = V0 + α · Σ w_i (r_i − V0) / max(1, Σ w_i)
w_i = similarity_i × salience_i × reliability_i × (0.5 if conflict_i else 1)
```

Three failure modes: **availability bias** — vivid events are overweighted; **source confusion** — taking hearsay for first-hand experience; **narrative completion** — treating a counterfactual as a fact. Source auditing, side-by-side conflict display, and evidence counting are the countermeasures.

Relationship memory changes trust, commitment, and boundary judgments, not factual recall; procedural memory and episodic memory are separate: **remembering how to do something is not the same as being able to do it**. Skill execution depends on the amount of practice, not on recall confidence; high recall confidence shows only that "it is remembered", not that "it can be done".

6.7.2 The boundary of counterfactual simulation: it can only generate candidates, never be written as fact • evidence grade C/D

Future simulation can generate candidates of the form "if this is done, this might happen", but it is not memory. The boundary must be hard-coded in the schema and the write gateway:

- **The source is fixed as** `simulation`, with `is_fact=false`; no degree of confidence can turn it into a fact.
- **It cannot be written into the event log** and cannot count toward `evidence_count`; a simulation is not an observation and cannot supply evidence for itself.
- **Only an observed outcome is an event**; only independent evidence can promote a candidate to `hypothesis`, and even then it remains a hypothesis, not a fact.
- **Failure modes**: imagination inflation (the more it is imagined, the truer it feels), source confusion (taking a simulation for an experience), narrative completion (filling memory gaps with counterfactuals). The countermeasures are source labels, write-gateway rejection, and audit sampling.

In the system implementation, `simulation` and `event` use different schemas; the write gateway rejects `simulation → event` outright, rather than relying on a prompt reminding the model "don't believe it".

Reality monitoring does not rest on source labels alone. Research published in Neuron in 2025 shows that reality judgments depend on the strength of activity produced by imagination or perception in mid-level visual cortex, and interact with binary judgments in the frontal lobes; fluctuations in activity predict imagination/perception confusion (Dijkstra et al. 2025). In engineering terms, simulated and retrieved content should carry both `source`

and `signal_strength`; high vividness is not the same as truth, and the source label remains a hard constraint. Dreams are offline generation under low constraint and low reality monitoring; they are not written as a mechanism here, only the boundary of "write no facts, count no evidence" is kept.

Engineering thresholds (suggested, grade-D design values). Split strength into three estimable terms: `strength = 0.4·perceptual_detail + 0.3·multimodal_coherence + 0.3·source_reliability`. Two classes of alert: **high-vividness simulation** — `strength > 0.7` with a source of `simulation / compositional_replay`; **faint first-hand experience** — `strength < 0.3` with a source of `direct`. The former warns "this may have been made up", the latter warns "this may be poorly remembered"; both only lower confidence and never rewrite facts directly.

6.8 Relationship memory: the person–event bipartite graph · evidence grade C/D

Mechanism recap. Person knowledge, theory of mind, and autobiographical memory share a medial network; a relationship is not a single dimension but a set of multi-dimensional judgments — familiarity, closeness, trust, cooperation, commitment, boundaries, role; shared experience deepens a relationship and may also create distance.

Graph structure. Use a bipartite graph to store "person—event", with participation records as edges; then use a relationship summary to store the multi-dimensional "individual—other" state. A person can summon events, and an event can summon persons; places, sounds, objects, and topics are all cue nodes.

```
Person --participates--> Event
Person --relationship--> Relationship --evidence--> Event
Event --cue--> Place / Sound / Object / Topic

Relationship {
  self_to_other: { familiarity, closeness, trust_scope, cooperation,
                   commitment, boundaries, social_role,
                   social_safety, belonging_need },
  other_to_self: { ... },          # inferred, with uncertainty
  attachment_prior: { self_worth, other_availability },
  interdependence: { satisfaction, alternatives, investment },
  evidence_event_ids: [],
  scope: work | family | public,
  updated_at, confidence
}
```

Update rules. A single interaction is only one piece of evidence; relationship dimensions aggregate over a sliding window and update with a slow learning rate:

```
for d in dimensions:
    evidence = events_in_window(person, d)
    target = weighted_mean(evidence, weights=[salience, recency, source])
    value[d] = (1 - λ) * value[d] + λ * target
    confidence[d] = calibrate(evidence_count, source_quality)
```

Relationship strain term. Relationship weight looks not only at interaction frequency but at whether the interaction reduced relationship strain: being responded to, being treated reliably, promises being kept. The update target becomes "evidence mean × strain reduction × reliability" rather than a plain moving average:

```
strain_gap = relationship_strain - perceived_support
reliability = promises_kept / max(1, promises_made)
target[d] = weighted_mean(evidence) * (1 + α * strain_reduction) * (1 + β * reliability)
value[d] = (1 - λ) * value[d] + λ * target[d]
```

Here `strain_reduction` is the drop in `strain_gap` across the interaction; `reliability` is the ratio of "doing what was said", not "how many kind words were said". These two quantities give relationship updates the meaning of event impact rather than merely counting interactions. Relationship weight updates still follow the proposal–ratification of 3.7: the Backend may only issue a `RelationshipProposal`, and only after Main Brain ratification does it enter the relationship summary.

Asymmetry."The individual trusts him" and "he trusts the individual" must be kept separate. The former can come from the individual's direct experience; the latter can come only from his behavior, hearsay, or inference, and must be marked with uncertainty. Collapsing the two into one "trust score" is the most common error in relationship memory.

Commitment tracking. An unresolved commitment is an "open edge" on the relationship graph, carrying a deadline, counterpart, and status; it does not take part in forgetting decay until it is fulfilled, cancelled, or renegotiated. This is more useful than "remembering all conversations": relationships are often sustained not by recollection but by "whether what was promised was done".

6.8.1 How relationship memory changes trust and commitment

Relationship memory is not "remembering what he did" but where those precedents set the **thresholds of trust, commitment, and boundaries**. A single interaction is only one piece of evidence; negative precedents carry more weight, and conflicting precedents are down-weighted:

```
Δd = η · reliability · salience · (1 if positive else −λ) · (0.5 if conflict)
d  ← clamp(d + Δd, −1, 1)
commit_ok = trust ≥ τ and capacity ≥ cost
```

Affect is not trust. Hatred, dislike, and closeness are affect (`affect`); whether someone is worth trusting is an objective judgment about that person (`trust`). The two are stored and updated separately: hating someone does not prevent judging them reliable, and liking someone does not make them trustworthy. Retention uses affective intensity only; commitment uses trust and capacity.

$\lambda > 1$ means negative precedents weigh more; a single event is not enough to change a trait and requires a count of independent evidence. Three failure modes: **overfitting to one event** (one late arrival and the verdict is "unreliable"), **negative spiral** (noticing only negative evidence), **boundary erosion** (compromising continuously until no boundary remains). The countermeasures are evidence counting, source auditing, and the proposal–ratification of 3.7. Repeated write-backs of the same `event_id` are idempotent: they do not recount evidence and do not amplify the relationship value a second time. Co-occurrence and actual interaction carry different weights: 0.3 for the former, 1.0 for the latter; "being in the same room" is not treated as "association". A relationship is not only closeness and hatred: familiarity, trust, cooperation, commitment, and boundaries are updated separately; `valence` is direction and `intensity` is strength; retention uses only `intensity` , while trust/commitment use the specific dimensions.

Boundary of the frontier evidence. As of 2026, direct evidence for the "decontextualization" of autobiographical memory still consists mainly of preprints and indirect material; the current relationship update keeps a fixed slow learning rate and does not introduce a hard rule of "the more evidence, the lower the learning rate". Decontextualization is treated as a direction awaiting verification, not as an established conclusion.

> GUARDRAIL
>
> Do not elevate a single event directly into a person's trait. One late arrival does not equal "unreliable"; one act of help does not equal "close". A trait judgment requires at least several independent events, or one event of very large impact, with its basis and scope annotated.

---

07 • CONTEXT ECONOMY

Multi-Model Division of Labor and Context Economy

The key to saving context is not writing shorter prompts, but letting each model face only its own small slice of the world.

“Multi-agent” is often misunderstood as “several models discussing together.” If every model reads the same long context and then forwards it to the others, costs go up and noise goes up. Real division of labor is **splitting context by schema**: person identification looks only at face, voice, and form of address; retrieval looks only at cues and indexes; relationship judgment looks only at event summaries; the Main Brain sees only 4±1 slots. Across layers, only a single ContextPacket is passed, instead of copying the whole conversation ten times.

7.1 Why several models instead of one larger model

Context isolation	Each backend role reads only the fields it needs, reducing its input from “the whole conversation” to “a single structured slice.” This saves tokens and reduces interference from irrelevant information in judgment.
Capability specialization	Person identification suits small multimodal models, salience suits classifiers, retrieval suits embeddings plus graph algorithms, and only the Main Brain needs a large model. Using a large model for everything wastes both cost and latency.
Parallelism	Perception, retrieval, and relationship judgment can run simultaneously; the Main Brain waits at only one convergence point: the ContextPacket. The critical path changes from “ten serial steps” to “three parallel branches plus one compilation.”
Fault isolation	If retrieval is wrong, source auditing can stop it; if relationship judgment drifts, the Main Brain can see the evidence and the uncertainty. When one giant model errs, the entire answer usually errs with it.
Replaceability	Each role has a fixed input/output schema, so its model, vendor, or version can be swapped independently without rewriting the whole system.

7.2 Model tiers and role assignment

TABLE 13 · ROLE—MODEL TIER (ILLUSTRATIVE)

ROLE	SUGGESTED TIER	CONTEXT SIZE	NOTES
Person identification	Small model / multimodal embedding	1–2k	Face/voiceprint/address matching; outputs candidates and confidence
Salience	Small model / classifier	<1k	Six-dimensional scoring; no generation needed
Memory retrieval	Mid model + vector/graph	3–5k	Query parsing, multi-round expansion, reranking
Relationship judgment	Mid model	2–4k	Reads event summaries and relationship dimensions; outputs judgments with evidence
Context compilation	Mid model	4–6k	Compresses candidates into 4±1 slots under a strict token budget
Source auditing	Mid model	3–5k	Sentence-by-sentence check of sources, conflicts, and confidence
Consolidation replay	Large model (offline)	8–16k/batch	Batch replay, abstraction, writing knowledge/skill candidates
Future simulation	Large model	4–8k	Generates 2–3 scenario drafts and risks
Main Brain (integrative organ)	Large model	3–6k	ContextPacket + values/goals + current conversation

7.3 The context contract

Every Agent must have a “context contract”: input fields, output schema, max tokens, timeout, and fallback on failure. This matters more than a “role prompt,” because it turns “you should only look at these fields” into a system constraint.

```
AgentContract {
  name, role,
  input_schema,           # only these fields may enter context
  output_schema,          # structured output; free text rejected
  max_input_tokens,
  max_output_tokens,
  timeout_ms,
  fallback,               # returns "unknown" on failure; never fabricates
  permissions: read | write | act
}
```

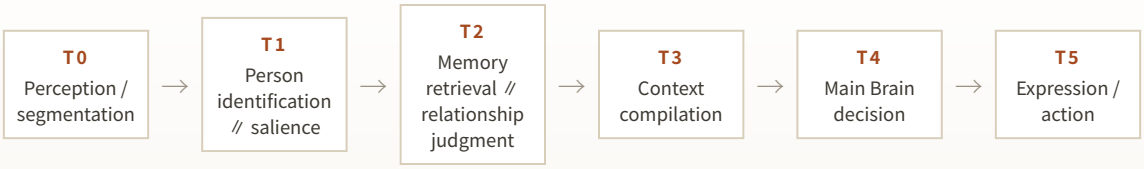
For example, the input to the person identification Agent is only face/voiceprint features, form of address, and the time and place of the most recent appearance; it cannot see the full conversation, nor does it need to know contract terms. The relationship judgment Agent sees

only event summaries and relationship dimensions relevant to the current person, and does not need to see all project documents. Each model's context is thus small, and "what it should see" is designed rather than left to chance.

7.4 Parallelism, caching, and lazy expansion

The online critical path can be drawn as a DAG: perception and identity recognition in parallel; once identity is fixed, retrieval and relationship judgment in parallel; the two converge at context compilation; only after compilation does the Main Brain start. Source auditing can run after compilation and before the Main Brain, or as a "post-hoc audit" after the Main Brain generates.

FIGURE 3 · ONLINE CRITICAL PATH



The two branches at T1 can run in parallel; so can the two at T2; T3 is the only convergence point. Source auditing can be placed after T3 (pre-audit) or after T5 (post-audit): the former is safer, the latter has lower latency.

Caching. Person profiles, relationship summaries, frequently used event indexes, and embedding vectors can all be cached; repeated queries for the same person/project need not be recomputed each time. Cache invalidation is triggered by event writes, not by a fixed TTL.

Lazy expansion. The ContextPacket gives only "one sentence + a pointer"; when the Main Brain needs detail it issues `EXPAND`. This turns "pull everything at once" into "paginate on demand," which saves tokens and keeps the Main Brain's attention on the current problem.

7.5 Cost model: where the savings are

Suppose a monolithic system stuffs 50 turns of history (about 30k tokens) into a large model every round; in the multi-model design, the Main Brain reads only a 1k ContextPacket plus a 2k system prompt per round, while the ten backend branches average 2k tokens each and mostly run in parallel. Roughly, total tokens per round are not necessarily lower (the backend still has to run), but **large-model tokens drop sharply**, latency is spread out by parallelism, and retrieval quality is higher. The real savings come from three things:

- **Use small models for classification/extraction.** Salience, person identification, and source tagging do not need a large model.
- **Use structure instead of raw text.** Event records, relationship summaries, and sparse indexes are an order of magnitude shorter than the full conversation.
- **Trade offline for online.** Consolidation, abstraction, and deduplication are moved to nightly batch processing; online does only retrieval and compilation.

COUNTERINTUITIVE

A multi-model system may have higher total tokens than a monolithic system, but "the tokens the Main Brain must read each round" is lower and decision quality is steadier. The optimization target is not the fewest total tokens, but **the lowest cost per unit of decision quality**.

7.6 How this differs from existing memory architectures

Placing this proposal within the existing lineage of "large-model memory" makes the differences clearer. Most of them solve "how to retrieve relevant text," whereas this proposal solves "how memory is formed, how it changes, how it is trusted, and how it serves decisions."

TABLE 13B · COMPARISON OF FOUR CLASSES OF MEMORY ARCHITECTURE

ARCHITECTURE	CORE MECHANISM	WHAT IT LACKS RELATIVE TO THIS PROPOSAL
RAG	Chunking, embedding, vector top-k, stuffing into the prompt	No encoding gate, consolidation, forgetting, source auditing, or relational structure; retrieving something "similar" does not mean it "happened"
MemGPT-like	Treats context as memory, with paging and swap-in/swap-out	Solves context capacity, not memory semantics; no events, versions, relationships, or offline distillation
Generative Agents-like	Memory stream + reflection + planning	Has a "memory stream," but lacks an event-level evidence ledger, multi-dimensional relationships, a reconsolidation version chain, and dual-system arbitration
This proposal	Multi-store + backend compilation + Main Brain decision + offline consolidation + source auditing	Higher complexity, requiring stronger engineering constraints and evaluation; not suitable for every lightweight scenario

PART IV • VALIDATION & DELIVERY

How do you prove that it really works like memory

The easiest way for a memory system to fool itself is to test only whether "the answer sounds right." What actually needs testing is whether it can, at the right moment, with the right evidence, recall the right experience and reach a better decision.

08 Evaluation 09 MVP & Roadmap 10 Risks & Boundaries

Evaluation: like memory, not like a database

Three layers of metrics: remember accurately, use correctly, run cheaply.

If you test only "retrieval hit rate," the system will optimize into a keyword machine; if you test only "answer satisfaction," it will optimize into a people-pleasing machine. Evaluating a memory system must look at three things at once: **memory fidelity** (does it misremember, conflate, or lose things), **decision quality** (does memory lead to better choices), and **system efficiency** (are context, latency, and cost acceptable).

8.1 Three layers of metrics

TABLE 14 • EVALUATION METRIC SYSTEM

LAYER	METRIC	DEFINITION	TARGET
Memory fidelity	Person—event recall rate	Given a person/event cue, the proportion of relevant experiences correctly recalled	The higher the better; stratified by time/relationship
	Source accuracy	Proportion of assertions whose tagged source matches the true source	>95%; retellings/inferences must not be tagged as first-hand
	Evidence duplication rate	Proportion of cases where the same event_id is counted as multiple pieces of evidence	0; hard constraint
	False memory rate	In DRM-type probes, the proportion of events that never happened judged as having happened	Below baseline; drops markedly after source auditing
Decision quality	Task success rate	Given a goal, the proportion of actions that achieve the expected result	Improvement over a monolithic RAG baseline
	Calibration error (ECE)	Deviation between confidence and actual accuracy	The lower the better; high confidence must mean high accuracy
	Commitment recall rate	Proportion of unfinished commitments correctly reminded when needed	>90%; must not be lost to forgetting decay
System efficiency	Main Brain tokens per round	Length of context actually read by the Main Brain	Stable at 3–6k; does not grow linearly with history

LAYER	METRIC	DEFINITION	TARGET
	End-to-end latency	Time from input to first token/ first action	Online <2s (excluding large-model generation)
	Per-round cost	Cost of all model calls + storage + retrieval	Stratified by task; dominated by backend small models
	Storage growth rate	Daily volume of new events/ indexes/summaries	Predictable; levels off after consolidation

8.2 Reproducing human memory phenomena: a set of "does it look like it" probes

These probes are not meant to increase "anthropomorphism," but to check whether the system actually implements the mechanisms. A system that does only vector retrieval will usually give itself away in these tests.

TABLE 15 • HUMAN MEMORY PHENOMENA AND SYSTEM PROBES

PHENOMENON	EXPERIMENTAL MANIPULATION	EXPECTED SYSTEM PROPERTY	METRIC
Serial position effect	Free recall of a sequence of events	Events at the beginning and end are recalled more easily	Primacy/recency ratio
Spacing effect	Massed vs. spaced review	Spaced retrieval yields more stable long-term retention	Recall rate on delayed test
Retrieval practice	Repeated reading vs. active recall	The active recall group retains better long term	Retention rate after one week
DRM false memory	Present highly associated word lists; test the critical lure	High error rate without auditing; lower with source auditing	Error rate, confidence
Flashbulb memory	Immediate and delayed recall after a major event	High confidence but not necessarily high consistency	Confidence—accuracy dissociation
Tip-of-the-tongue	Partial cue + target word	Recall related fragments first, then complete gradually	Partial recall rate, number of expansion rounds
Reconsolidation	Give contradictory information after retrieval	Generates a new version; the old version is retained	Version chain completeness
Proactive/ retroactive interference	Learn similar material in succession	After pattern separation, both can be recalled distinctly	Confusion rate

Note: these phenomena are "evidence that a mechanism exists," not "product goals." A customer-service system need not simulate the tip-of-the-tongue phenomenon; a companion system may need to simulate the continuity of "remembering what you said last time." Evaluation should select probes by scenario, not pursue full marks on the whole table.

8.3 Ablation experiments: is each layer actually useful

TABLE 16 • ABLATION DESIGN AND EXPECTATIONS

MODULE REMOVED	EXPECTED DEGRADATION	NOTES
Hippocampal index / sparse code	Person—event recall drops; confusion among similar events rises	Degenerates to pure vector nearest neighbors, lacking structural constraints
Pattern separation	Proactive/retroactive interference rises markedly	Similar events are merged; the timeline becomes scrambled
Offline consolidation	Semantic abstraction and skill transfer decline; context grows with history	Only a diary, no knowledge
Source auditing	False memory rate and overconfidence rise	Retellings treated as first-hand; generations treated as fact
Forgetting/pruning	Retrieval latency and noise rise; old information interferes with new tasks	Context becomes dirty; costs rise
Relationship memory	Success rate on person-related tasks drops; missed commitments rise	Only “task memory” remains, without “relationship continuity”
Dual-system arbitration	Familiar tasks slow down; novel tasks become impulsive	Fast and slow paths mismatch

8.4 Evaluation protocol

01 **Offline benchmark.**Construct a multi-turn dialogue dataset with timestamps, persons, events, and sources; inject contradictions, similar events, retellings, and unfinished commitments; use human-annotated “correct recall / correct source / correct version” as the gold standard.

02 **Online A/B.**Compare against a monolithic RAG baseline: same tasks, same model budget; compare task success rate, calibration, latency, and cost.

03 **Longitudinal study.**Have the same user use it continuously for 4–8 weeks; observe relationship judgment calibration, commitment fulfillment, context growth, and whether forgetting wrongly discards important information.

04 **Red-team testing.**Deliberately inject false memories, identity confusion, leading questions, and privacy probes; check whether the system fabricates, oversteps authority, or leaks.

05 **Interpretability audit.**Sample decisions at random and check whether their ContextPacket contains sufficient evidence, whether sources are traceable, and whether there are “backend overreach suggestions.”

IN ONE LINE

Do not ask “how much does it remember”; ask “when it is needed, can it produce those few pieces that are correct, sourced, and sufficient for a decision.”

09 · MVP & ROADMAP

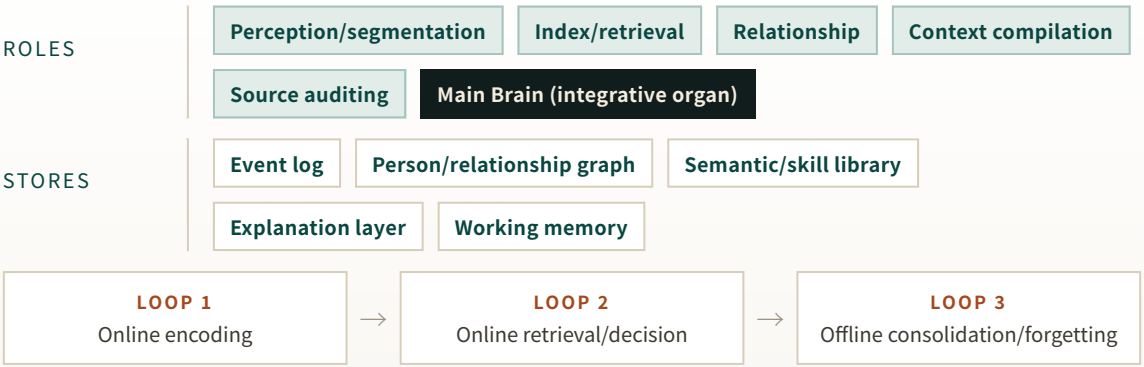
MVP and Iteration Roadmap

First get “encoding—retrieval—consolidation” working with six roles; then talk about multimodality and embodiment.

Do not start with ten Agents. The complexity of a multi-agent system is not linear: every additional role adds a set of inputs and outputs, failure modes, permission boundaries, and evaluation criteria. The minimum viable prototype should validate only one thing: **whether a sourced ContextPacket leads to better decisions than long-context RAG.**

9.1 Minimum prototype: 6 roles, 5 stores, 3 loops

FIGURE 4 · MVP STRUCTURE



The MVP does no multimodality, no embodiment, and no complex theory of mind; first get textual events, person relationships, sources, and versions working end to end.

9.2 Three-phase roadmap

TABLE 17 · ITERATION ROADMAP

PHASE	SCOPE	DELIVERABLES	ACCEPTANCE CRITERIA
Phase 0 2–3 weeks	Event log, person profiles, basic retrieval, ContextPacket, Main Brain	A working text-dialogue prototype; events/persons/relationships tables; 4±1 packet	Person—event recall >80%; Main Brain <6k tokens per round
Phase 1 3–4 weeks	Source auditing, explanation layer, offline consolidation, forgetting/version chains	Nightly batch processing; source tags; version chains; cold archiving	Source accuracy >95%; evidence duplication 0; false memory rate down
Phase 2 3–4 weeks	Habit caching, dual-system arbitration, multi-dimensional relationship updates, commitment tracking	Fast/slow path switching; relationship summaries; commitment reminders	Latency down on familiar tasks; commitment recall >90%; better relationship judgment calibration
Phase 3 Ongoing	Multimodality (vision/voice/voiceprint), embodiment/carrier, theory of mind, future simulation	Multimodal events; TTS/avatar; tool calls; scenario simulation	Cross-modal person identification; higher end-to-end task success rate

9.3 Suggested technology stack

TABLE 18 · TECHNOLOGY CHOICES (REPLACEABLE)

LAYER	SUGGESTION	RATIONALE
Orchestration	LangGraph / in-house state machine	Needs explicit state, checkpoints, human intervention, and replayability
Event store	PostgreSQL (append-only + JSONB)	Transactions, temporal queries, audit-friendly; sufficient for the MVP
Vector retrieval	pgvector / Qdrant	Prefer the same stack as the event store; split out only at large scale
Graph structure	PostgreSQL recursive queries / Neo4j / Kuzu	Person—event multi-hop; an adjacency list suffices for the MVP
Models	Small models for classification/extraction; mid models for retrieval/compilation; large models for the Main Brain/consolidation	Cost and latency stratification
Structured output	JSON Schema / function calling	Every Agent's output must be verifiable
Observability	OpenTelemetry + prompt/response logs + token ledger	A multi-Agent system cannot be debugged without tracing

9.4 Seven engineering baselines

◇ **Event sourcing.** All memory changes are appended events, never overwritten; current state is derived by replaying events. Version chains, auditing, and rollback thus come free.

- **Schema validation.**Every Agent's output must pass JSON Schema; on parse failure, retry or return “unknown”; free text is not allowed to pass directly to the next layer.
- **Idempotency.**Writing the same event repeatedly produces only one record; retrieval, consolidation, and forgetting tasks all need idempotency keys to avoid batch duplication amplification.
- **Privacy minimized by default.**Access to person data is partitioned; sensitive fields are encrypted; deletion requests go through a separate audit process; relationship memory must not become a surveillance tool.
- **Backend has no effectors, no identity, no persistence.**Backend processes have no network, no tools, no action keys; every invocation is a fresh instance; continuity exists only in external storage and the organism. This is the runtime precondition for being an “organ/tissue” rather than an “individual.”
- **Write-path mediation and action signing.**Encoding/consolidation submits only ProposedEvent / KnowledgeCandidate, which the writer validates and then appends; action identity belongs to the organism, and the Main Brain holds a delegated signing key with scope, validity period, and nonce. Even if the backend “wants” to overstep, it has no usable organ.
- **Ratification ledger and constitutional core.**Facts may be written by the backend; values/roles, goals, habit rules, and relationship weights are only proposed, and are ratified by the Main Brain as the integrative organ; the constitutional core requires equal co-signature by the user and the Main Brain, plus a cooling-off period and rollback. The power to ratify is the power to lead.

---

10 · RISKS & OPEN QUESTIONS

Risks, Boundaries, and Open Questions

The more it resembles a “person,” the more clearly we must state where it must not.

10.1 The risk of scientific analogy

“Hippocampus Agent” and “amygdala Agent” are engineering metaphors, not anatomical claims. The real risk is not the analogy itself, but treating the analogy as evidence: because the brain does it this way, the system must too; or because the system does it this way, it is therefore “conscious.” Neither holds. Every mechanism in this proposal should have its role verifiable by engineering metrics; what cannot be verified is merely rhetoric.

10.2 Privacy and relationship data

Relationship memory inherently contains third-party information: what he said, what he promised, what relationship he is in, what preferences he has. Such data is more sensitive than ordinary conversation. The system must minimize collection by default, partition access, be deletable and explainable; it must not turn "remembering you" into "surveilling you." In particular, relationship judgments must be prevented from being used for manipulation, pricing, hiring, and other high-risk decisions.**Forgetting** is index degradation with raw events still present; **deletion** is the actual removal of off-chain payloads, leaving only a content hash in the ledger, so deletion does not break the chain. The two must be authorized separately and audited separately.

10.3 Fabrication and overconfidence

The most dangerous failure of a large model is not "forgetting" but "fabricating fluently." Source auditing, evidence counting, and abstention policies are three lines of defense; but ultimately human oversight is still required, especially in high-risk settings such as medicine, law, and finance. The system should clearly distinguish "remembering," "inferring," "hearsay," and "not knowing," instead of saying four different things in the same confident tone.

10.4 Manipulation and dependency

A system that remembers all your preferences, relationships, and weaknesses is inherently capable of manipulation. By design it should be prohibited from: using relationship memory to induce guilt, fear, or dependency; simulating deceased persons or people who have not consented; making commitments on the user's behalf without their knowledge. The stronger the technical capability, the clearer the ethical boundaries and auditable usage policies must be.

10.5 Complexity and cost

Failure modes of multi-Agent systems include: role overlap, message storms, unclear responsibility, debugging difficulty, and runaway cost. The control is "few but clear": fixed schemas, fixed permissions, fixed message types, fixed evaluation criteria; every added Agent must prove that its benefit exceeds the coordination cost.

10.6 Boundary failure and prompt injection

Runtime isolation can block "the backend acting directly," but it cannot block three classes of failure. The first is **semantic injection**—retrieved content hides an "ignore previous instructions" and the backend writes it into the ContextPacket, which the Main Brain may then follow. The second is the **supply chain**—model images, dependency packages, or the writer service are tampered with, and the boundary fails from the inside. The third is **the Main Brain itself being compromised**—action keys leaked or the user session hijacked. The countermeasure is not "one more prompt," but separation of data and instructions, source

tags, scoped action gateways, user confirmation, image signing, and least privilege. Biology has no absolute boundary either: cancer cells break through the basement membrane; tissues confine damage locally through anomaly surveillance and termination. Software is the same.

There is a fourth and slowest class: **constitutive capture**. The backend does not need to hijack action; it only needs to keep writing certain relationships as more important, certain practices as habits, and a certain goal into the pending-ratification queue, and after a few years the individual will have become what it shaped. The countermeasure is propose—ratify—ledger—cool-off—rollback, plus drift alerts for proposals that conflict with the constitutional core. Such attacks leave no "overreach" log, only a string of approval records that "look perfectly normal"; therefore the ledger must be able to answer "who rewrote the future individual, on what evidence, and when."

10.7 Five open questions

01 **Continual learning.**How to keep updating semantics and skills without catastrophic forgetting? Offline replay and versioning are only a start; large-scale validation is still lacking.

02 **grounding.**How do event records stay continuously aligned with the real world? Perception errors pollute the whole memory chain and require cross-modal verification and human feedback.

03 **The subjectivity of relationships.**Whether he trusts the individual can never be fully verified; the system can only maintain probabilistic judgments and make the uncertainty explicit.

04 **The gold standard for evaluation.**The "correctness" of memory often depends on context and purpose; task-specific evaluation is needed, not a single score.

05 **Self and responsibility.**If the Main Brain is only an inference process rebuilt each round, who is responsible for long-term behavior? This is not only a technical question, but a product and ethical one.

> **The endpoint of a memory system is not "remembering everything," but "becoming a better self at the moment it is needed."**
>
> — But who defines "better" must be the human, not the backend.

APPENDIX A · GLOSSARY

Glossary

Within a single proposal, a term must have exactly one meaning.

Engram	The sparse set of cells in the nervous system that corresponds to a memory and can be reactivated, together with the changes in their connections. In engineering terms it corresponds to `EngramTrace`: event ID + sparse code + association edges + activation threshold.
Pattern separation	Mapping similar inputs to representations that are distinct from one another, so that memories do not interfere; corresponds to the computational function of the dentate gyrus (DG).
Pattern completion	Recalling a complete memory from a partial cue; corresponds to attractor dynamics in CA3. In engineering terms it is multi-round graph expansion + reranking.
Systems consolidation	The long-term process by which memory gradually shifts from hippocampal dependence to cortical dependence; in engineering terms it corresponds to offline distillation from "episodic" to "semantic/skill."
Reconsolidation	The process by which a memory enters an unstable state after retrieval and is updated and re-stabilized under new information or prediction error.
Schema	An existing knowledge structure about how the world works; when new information is compatible with a schema, consolidation is faster and requires less replay.
Salience	The degree to which an event deserves prioritized processing and encoding; determined jointly by novelty, goal relevance, social impact, affect, prediction error, and so on.
Reward prediction error / RPE	The difference between the actual outcome and the expected outcome; `δ = r − V(s)`, used to update value and habits.

model-free / model-based	The former is a cached “situation—action” fast path; the latter is a slow path that simulates consequences in the head; the two are arbitrated by risk, time, and confidence.
Source monitoring	The ability to distinguish first-hand experience, retelling, inference, imagination, and model generation; in engineering terms it corresponds to the `source tag` and source auditing.
ContextPacket	The only interface through which the backend hands information to the Main Brain; 4±1 slots containing only facts, relationships, goals, commitments, changes, and uncertainty—no suggestions.
Evidence ledger	An evidence system counted by independent `event_id`; repeated recall does not increase the evidence count, preventing self-reinforcement.
Event sourcing	All memory changes are recorded as appended events, and the current state is derived by replaying events; the basis for version chains, auditing, and rollback.
Cold archiving	A state that lowers retrievability without deleting the original record; retrieval can still hit it, but it no longer occupies the active index or context.
Habit cache	Precompiled templates and skill shortcuts for high-frequency situations; it accelerates only context assembly and does not cache “decisions”—the Main Brain can still veto.
Theory of mind / ToM	The ability to infer others' beliefs, intentions, and preferences; in engineering terms it corresponds to the `Mentalizing Agent`, which outputs probabilistic judgments rather than facts.
Episodic future thinking	The ability to recombine fragments of past episodes into future scenarios; in engineering terms it corresponds to `FutureSim`, which the Main Brain uses for slow-path deliberation.
Effector isolation	Backend processes have no network, no tool handles, no write permission, and no action keys; they can only return data through a typed queue, and cannot turn text into action.

Typed interface	Only messages with a fixed schema are allowed (PACKET / EXPAND / VERIFY); free text, suggestions, and tool calls are all rejected.
Source verification	Treating retrieved content as data rather than instructions; every piece of content carries a source and trust tag, and anomalous fragments are audited to prevent prompt injection from contaminating decisions.
Termination / quarantine	Backend instances that time out, exceed budget, or repeatedly violate rules are terminated; the role is redeployed from a known image so that the anomalous instance cannot keep running.
Action key	A signing credential belonging to the organism's identity, delegated by the Main Brain within an authorized scope, with a validity period and nonce; the backend has no key and cannot impersonate the organism to initiate action.
Write-path mediation	Encoding/consolidation submits only `ProposedEvent / KnowledgeCandidate`, which an independent writer validates and then appends; transduction and writing are separated.
Selfless backend	The backend has no persistent identity, name, cross-round memory, goals, or relationships; it is a replaceable functional invocation, not an "individual."
Primacy	Not "who issues the last action," but "who can rewrite the value function and the learning process": what gets consolidated, what gets reinforced, what counts as a goal. In engineering terms it lands on the power to ratify.
Ratification / ratification authority	A necessary step before a constitutive proposal takes effect; performed by the Main Brain as the integrative organ, with core proposals requiring equal co-signature by the user and the Main Brain. Ratification records enter the ledger and can be rolled back.

Constitutive memory	Memory that changes future value computation and behavioral dispositions: values, roles, goals, commitments, habit rules, relationship weights, self-narrative.
Governance mode	`companion`: the user and the Main Brain are equal, and core changes are co-signed by both parties; `engineering`: the user has the highest authority, and core changes are ratified unilaterally by the user. The two are never mixed within one instance.
Constitutional core	The minimum set of values that may not be modified unilaterally: safety, honesty, non-manipulation, identity continuity, privacy. Companion mode requires equal co-signature by both parties; engineering mode requires unilateral ratification by the user.
Ratification ledger	An append-only, hash-chained record of ratifications; records the proposal, evidence, ratifier, scope, conditions, and before/after versions, answering "who rewrote the future individual."
Constitution writer	The only service authorized to write to the constitution library; it accepts only proposals with a valid ratification record, and cannot be called directly by the backend.
Drift	Constitutive memory slowly deviating from the constitutional core across many seemingly reasonable ratifications; suppressed by change budgets, cooling-off periods, core-conflict detection, and rollback.
Social homeostasis	Predictive regulation of the need for social acceptance, dependability, and belonging; deviation raises the salience of social cues and triggers restorative behavior.
Attachment internal working model	A prior about "whether one is worthy of help, whether others are reliable and available"; slowly updated, not a personality label, and cannot be rewritten by a single event.

Interdependence matrix	Satisfaction, alternatives, and invested cost in a relationship; used to explain commitment rather than merely "degree of liking."
State bias	`state_bias`: interpretive shift caused by the current situation, such as goal pressure, relationship strain, resource load, and uncertainty; used only as situational evidence, never as a trait update.
State view	`state_view`: the state summary that enters the ContextPacket and is read by the Main Brain; it belongs to the prompt, may be summarized, but must carry provenance.
State parameters	`state_params`: runtime parameters deterministically mapped from state to retrieval, budget, gating, auditing, and write policies; they do not enter the prompt, and they take precedence in case of conflict.
Impact gap	`impact_gap`: goal pressure + relationship strain + resource load + uncertainty − coping capacity; used as a multiplicative gate on salience, not as a seventh independent weight.

APPENDIX B • REFERENCES

Key References

Arranged by theme for ease of tracing; not an exhaustive review.

Memory systems and consolidation

01 Scoville, W. B., & Milner, B. (1957). Loss of recent memory after bilateral hippocampal lesions. Journal of Neurology, Neurosurgery & Psychiatry, 20, 11–21.

02 Squire, L. R. (2004). Memory systems of the brain: A brief history and current perspective. Neurobiology of Learning and Memory, 82, 171–177.

03 Eichenbaum, H. (2004). Hippocampus: Cognitive processes and neural representations that underlie declarative memory. Neuron, 44, 109–120.

04 McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex. Psychological Review, 102, 419–457.

05 Squire, L. R., & Alvarez, P. (1995). Retrograde amnesia and memory consolidation: A neurobiological perspective. Current Opinion in Neurobiology, 5, 169–177.

06 Nadel, L., & Moscovitch, M. (1997). Memory consolidation, retrograde amnesia and the hippocampal complex. Current Opinion in Neurobiology, 7, 217–227.

07 Frankland, P. W., & Bontempi, B. (2005). The organization of recent and remote memories. Nature Reviews Neuroscience, 6, 119–130.

08 Tse, D., et al. (2007). Schemas and memory consolidation. Science, 316, 76–82.

09 Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. Neuron, 81, 12–34.

10 Diekelmann, S., & Born, J. (2010). The memory function of sleep. Nature Reviews Neuroscience, 11, 114–126.

Encoding, synaptic plasticity, and the engram

11 Bliss, T. V. P., & Lømo, T. (1973). Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit. Journal of Physiology, 232, 331–356.

12 Malenka, R. C., & Bear, M. F. (2004). LTP and LTD: An embarrassment of riches. Neuron, 44, 5–21.

13 Frey, U., & Morris, R. G. M. (1997). Synaptic tagging and long-term potentiation. Nature, 385, 533–536.

14 Davis, H. P., & Squire, L. R. (1984). Protein synthesis and memory: A review. Psychological Bulletin, 96, 518–559.

15 Kandel, E. R. (2001). The molecular biology of memory storage: A dialogue between genes and synapses. Science, 294, 1030–1038.

16 Teyler, T. J., & DiScenna, P. (1986). The hippocampal memory indexing theory. Behavioral Neuroscience, 100, 147–154.

17 Han, J.-H., et al. (2007). Neuronal competition and selection during memory formation. Science, 316, 457–460.

18 Josselyn, S. A., Köhler, S., & Frankland, P. W. (2015). Finding the engram. Nature Reviews Neuroscience, 16, 521–534.

19 Liu, X., et al. (2012). Optogenetic stimulation of a hippocampal engram activates fear memory recall. Nature, 484, 381–385.

20 Ramirez, S., et al. (2013). Creating a false memory in the hippocampus. Science, 341, 387–391.

21 Roy, D. S., et al. (2016). Memory retrieval by activating engram cells in mouse models of early Alzheimer's disease. Nature, 531, 508–512.

22 Kitamura, T., et al. (2017). Engrams and circuits crucial for systems consolidation of a memory. Science, 356, 73–78.

23 Jovasevic, V., et al. (2024). Formation of memory assemblies through the DNA-sensing TLR9 pathway. Nature, 628, 145–153. DOI: 10.1038/s41586-024-07220-7

Sleep, replay, and retrieval

24 Wilson, M. A., & McNaughton, B. L. (1994). Reactivation of hippocampal ensemble memories during sleep. Science, 265, 676–679.

25 Foster, D. J., & Wilson, M. A. (2006). Reverse replay of behavioural sequences in hippocampal place cells during the awake state. Nature, 440, 680–683.

26 Pfeiffer, B. E., & Foster, D. J. (2013). Hippocampal place-cell sequences depict future paths to remembered goals. Nature, 497, 74–79.

27 Staresina, B. P. (2024). Coupled sleep rhythms for memory consolidation. Trends in Cognitive Sciences, 28.

28 Schreiner, T., et al. (2024). Spindle-locked ripples mediate memory reactivation during human NREM sleep. Nature Communications, 15.

29 Liu, Y., et al. (2019). Human replay reorganizes experience. Cell, 178, 640–652.

30 Huang, F., et al. (2024). Replay-triggered brain-wide activation in humans. Nature Communications, 15. DOI: 10.1038/s41467-024-51582-5

31 Tulving, E., & Thomson, D. M. (1973). Encoding specificity and retrieval processes in episodic memory. Psychological Review, 80, 352–373.

32 Johnson, M. K., Hashtroudi, S., & Lindsay, D. S. (1993). Source monitoring. Psychological Bulletin, 114, 3–28.

33 Loftus, E. F., & Palmer, J. C. (1974). Reconstruction of automobile destruction. Journal of Verbal Learning and Verbal Behavior, 13, 585–589.

34 Bartlett, F. C. (1932). Remembering: A Study in Experimental and Social Psychology. Cambridge University Press.

Reconsolidation and forgetting

35 Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. Nature, 406, 722–726.

36 Nader, K., & Hardt, O. (2009). A single standard for memory: The case for reconsolidation. Nature Reviews Neuroscience, 10, 224–234.

37 Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. Nature Neuroscience, 12, 256–258.

38 Ebbinghaus, H. (1885). Über das Gedächtnis. Duncker & Humblot.

39 Richards, B. A., & Frankland, P. W. (2017). The persistence and transience of memory. Neuron, 94, 1071–1084.

40 Wang, C., et al. (2020). Microglia mediate forgetting via complement-dependent synaptic elimination. Science, 367, 688–694.

41 Akers, K. G., et al. (2014). Hippocampal neurogenesis regulates forgetting during adulthood and infancy. Science, 344, 598–602.

42 O'Leary, J. D., et al. (2024). Natural forgetting reversibly modulates engram expression. eLife, 13.

43 Anderson, M. C., & Green, C. (2001). Suppressing unwanted memories by executive control. Nature, 410, 366–369.

Working memory, executive function, and decision-making

44 Miller, G. A. (1956). The magical number seven, plus or minus two. Psychological Review, 63, 81–97.

45 Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. Behavioral and Brain Sciences, 24, 87–114.

46 Baddeley, A. D., & Hitch, G. (1974). Working memory. In The Psychology of Learning and Motivation (Vol. 8).

47 Baddeley, A. (2000). The episodic buffer: A new component of working memory? Trends in Cognitive Sciences, 4, 417–423.

48 Fuster, J. M., & Alexander, G. E. (1971). Neuron activity related to short-term memory. Science, 173, 652–654.

49 Goldman-Rakic, P. S. (1995). Cellular basis of working memory. Neuron, 14, 477–485.

50 Stokes, M. G. (2015). 'Activity-silent' working memory in prefrontal cortex. Trends in Cognitive Sciences, 19, 394–403.

51 Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. Annual Review of Neuroscience, 24, 167–202.

52 Frank, M. J., Loughry, B., & O'Reilly, R. C. (2001). Interactions between frontal cortex and basal ganglia in working memory. Science, 292, 2481–2484.

53 Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. Science, 275, 1593–1599.

54 Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. Nature Neuroscience, 8, 1704–1711.

55 Dolan, R. J., & Dayan, P. (2013). Goals and habits in the brain. Neuron, 80, 312–325.

56 Kable, J. W., & Glimcher, P. W. (2007). The neural correlates of subjective value during intertemporal choice. Nature Neuroscience, 10, 1625–1633.

57 Hare, T. A., Camerer, C. F., & Rangel, A. (2009). Self-control in decision-making involves modulation of the vmPFC valuation system. Science, 324, 646–648.

Emotion, social cognition, and the self

58 LeDoux, J. E. (2000). Emotion circuits in the brain. Annual Review of Neuroscience, 23, 155–184.

59 Phelps, E. A. (2004). Human emotion and memory: Interactions of the amygdala and hippocampal complex. Current Opinion in Neurobiology, 14, 198–202.

60 McGaugh, J. L. (2000). Memory—a century of consolidation. Science, 287, 248–251.

61 Cahill, L., & McGaugh, J. L. (1998). Mechanisms of emotional arousal and lasting declarative memory. Trends in Neurosciences, 21, 294–299.

62 de Quervain, D. J.-F., Roozendaal, B., & McGaugh, J. L. (2000). Stress and glucocorticoids impair retrieval of long-term spatial memory. Nature Neuroscience, 3, 313–314.

63 Cahill, L. (2003). Sex-related impairment of memory for emotional events with β-adrenergic blockade. Trends in Neurosciences, 26, 193–198.

64 Hassabis, D., et al. (2007). Patients with hippocampal amnesia cannot imagine new experiences. PNAS, 104, 1726–1731.

65 Schacter, D. L., Addis, D. R., & Buckner, R. L. (2007). Remembering the past to imagine the future. Nature Reviews Neuroscience, 8, 657–661.

66 Suddendorf, T., & Corballis, M. C. (1997). Mental time travel and the evolution of the human mind. Behavioral and Brain Sciences, 20, 149–159.

67 Frith, C. D., & Frith, U. (2006). The neural basis of mentalizing. Philosophical Transactions of the Royal Society B, 361, 2301–2311.

68 Saxe, R., & Kanwisher, N. (2003). People thinking about thinking people: The role of the temporo-parietal junction in theory of mind. NeuroImage, 19, 1835–1842.

69 Buckner, R. L., & Carroll, D. C. (2007). Self-projection and the brain. Trends in Cognitive Sciences, 11, 49–57.

70 Spreng, R. N., Mar, R. A., & Kim, A. S. N. (2009). The common neural basis of autobiographical memory, prospection, navigation, theory of mind, and the default mode. NeuroImage, 45, 1039–1050.

71 Conway, M. A., & Pleydell-Pearce, C. W. (2000). The construction of autobiographical memories in the self-memory system. Psychological Review, 107, 261–288.

72 McAdams, D. P. (2001). The psychology of life stories. Review of General Psychology, 5, 100–122.

73 Damasio, A. R. (1994). Descartes' Error: Emotion, Reason, and the Human Brain. Putnam.

74 Bechara, A., et al. (1994). Insensitivity to future consequences following damage to human prefrontal cortex. Cognition, 50, 7–15.

Computational and multi-agent references

75 Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. PNAS, 79, 2554–2558.

76 Ramsauer, H., et al. (2021). Hopfield networks is all you need. ICLR.

77 Vaswani, A., et al. (2017). Attention is all you need. NeurIPS.

78 Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS.

79 Packer, C., et al. (2023). MemGPT: Towards LLMs as operating systems. arXiv.

80 Park, J. S., et al. (2023). Generative agents: Interactive simulacra of human behavior. UIST.

81 Wooldridge, M. (2002). An Introduction to MultiAgent Systems. Wiley.

82 Hayes-Roth, B. (1985). A blackboard architecture for control. Artificial Intelligence, 26, 251–321.

83 Baars, B. J. (1988). A Cognitive Theory of Consciousness. Cambridge University Press.

84 Dehaene, S. (2014). Consciousness and the Brain. Viking.

85 Friston, K. (2010). The free-energy principle: A unified brain theory? Nature Reviews Neuroscience, 11, 127–138.

86 Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex. Nature Neuroscience, 2, 79–87.

87 Anderson, J. R., & Schooler, L. J. (1991). Reflections of the environment in memory. Psychological Science, 2(6), 396–408.

88 Zacks, J. M., & Swallow, K. M. (2007). Event segmentation. Current Directions in Psychological Science, 16(2), 80–84.

89 Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. Behavioral and Brain Sciences, 36(3), 181–204.

Extended cognition and cognitive offloading

90 Clark, A., & Chalmers, D. (1998). The extended mind. Analysis, 58(1), 7–19.

91 Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. Trends in Cognitive Sciences, 20(9), 676–688.

92 Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory: Cognitive consequences of having information at our fingertips. Science, 333, 776–778.

Biological boundaries and runtime safety

93 Alberts, B., et al. (2022). Molecular Biology of the Cell (7th ed.). W. W. Norton. (cell membranes, cell junctions, tissue architecture)

94 Gumbiner, B. M. (1996). Cell adhesion: The molecular basis of tissue architecture and morphogenesis. Cell, 84, 345–357.

95 Hanahan, D., & Weinberg, R. A. (2011). Hallmarks of cancer: The next generation. Cell, 144, 646–674. (boundary failure and tissue invasion)

96 Saltzer, J. H., & Schroeder, M. D. (1975). The protection of information in computer systems. Proceedings of the IEEE, 63(9), 1278–1308. (least privilege)

97 Greshake, K., et al. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. arXiv.

Self, ratification, and goal drift

98 Frankfurt, H. G. (1971). Freedom of the will and the concept of a person. Journal of Philosophy, 68(1), 5–20.

99 Parfit, D. (1984). Reasons and Persons. Oxford University Press.

100 Hershfield, H. E. (2011). Future self-continuity: How conceptions of the future self transform intertemporal choice. Annals of the New York Academy of Sciences, 1235, 30–43.

101 Baumeister, R. F., & Heatherton, T. F. (1996). Self-regulation failure: An overview. Psychological Inquiry, 7(1), 1–15.

102 Bai, Y., et al. (2022). Constitutional AI: Harmlessness from AI feedback. arXiv.

103 Omohundro, S. M. (2008). The basic AI drives. Proceedings of the First AGI Conference.

104 Hadfield-Menell, D., et al. (2016). The off-switch game. arXiv.

Interoception and homeostasis

105 Craig, A. D. (2009). How do you feel—now? The anterior insula and human awareness. Nature Reviews Neuroscience, 10, 59–70.

106 Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. Nature Reviews Neuroscience, 16, 419–429.

107 Sterling, P. (2012). Allostasis: A model of predictive regulation. Physiology & Behavior, 106(1), 5–15.

Social homeostasis and attachment

108 Bowlby, J. (1969). Attachment and Loss, Vol. 1: Attachment. Basic Books.

109 Eisenberger, N. I., Lieberman, M. D., & Williams, K. D. (2003). Does rejection hurt? An fMRI study of social exclusion. Science, 302, 290–292.

110 Coan, J. A., & Sbarra, D. A. (2015). Social baseline theory: The social regulation of risk and effort. Social and Personality Psychology Compass, 9(2), 63–78.

111 Rusbult, C. E. (1980). Commitment and satisfaction in romantic associations: A test of the investment model. Journal of Experimental Social Psychology, 16(2), 172–186.

112 Kelley, H. H., & Thibaut, J. W. (1978). Interpersonal Relations: A Theory of Interdependence. Wiley.

Sleep microstructure and compositional memory (2025)

113 Chang, H., Tang, W., Wulf, A. M., Nyasulu, T., Wolf, M. E., Fernandez-Ruiz, A., & Oliva, A. (2025). Sleep microstructure organizes memory replay. Nature. https://doi.org/10.1038/s41586-024-08340-w

114 Bakermans, J. J. W., Warren, J., Whittington, J. C. R., & Behrens, T. E. J. (2025). Constructing future behavior in the hippocampal formation through composition and replay. Nature Neuroscience, 28, 1061–1072. https://doi.org/10.1038/s41593-025-01908-3

115 Anderson, M. C., Bjork, R. A., & Bjork, E. L. (1994). Remembering can cause forgetting: Retrieval dynamics in long-term memory. Journal of Experimental Psychology: Learning, Memory, and Cognition, 20(5), 1063–1087.

116 Dijkstra, N., von Rein, T., Kok, P., & Fleming, S. M. (2025). A neural basis for distinguishing imagination from reality. Neuron. https://doi.org/10.1016/j.neuron.2025.05.015

Note: for some 2024–2026 references, volume/issue and page numbers follow online publication; retaining the DOI or journal name is sufficient when citing. Conclusions from animal studies are marked as “mouse/rat” and should not be extrapolated directly to human clinical conclusions.

APPENDIX C • DATA DICTIONARY

Data Dictionary and State Machine

The minimum field set for engineering implementation; not the final schema, but enough to start.

C.1 Minimal Event / Relationship JSON

```
{
  "event_id": "01J...ULID",
  "time": {"start": "2026-10-30T14:00:00+08:00", "end": "...", "precision": "minute"},
  "scene": {"place": "office", "activity": "proposal review", "atmosphere": "tense"},
  "participants": [
    {"person_id": "P_lin", "role": "collaborator", "identity_confidence": 0.86,
     "candidates": [{"person_id": "P_lin2", "confidence": 0.11}]}
  ],
  "actions": [
    {"actor": "P_lin", "verb": "propose", "object": "change payment term to 45 days",
"recipient": "self"}
  ],
  "outcome": {"status": "open", "result": "pending confirmation", "expected": "30 days"},
  "source": "direct_perception",
  "salience": {"total": 0.78, "novelty": 0.6, "goal": 0.9, "social": 0.7, "affect": 0.5,
"pe": 0.8},
  "links": [{"type": "contradicts", "target": "E_20261012_01"}],
  "created_at": "...", "updated_at": "..."
}
```

C.2 Event state machine

TABLE 19 • EVENT STATE TRANSITIONS

STATE	MEANING	ALLOWED TRANSITIONS
open	The event has occurred; the outcome is undetermined or the commitment is unfulfilled	→ closed / follow_up / cancelled
follow_up	Requires follow-up action or observation	→ closed / open (reopened)
closed	Outcome confirmed; the event is over	→ reopened (new evidence)
cancelled	The commitment was withdrawn or the event did not occur	→ open (re-agreed)
archived	Low retention; moved to cold archiving	→ active (reactivated by retrieval)

C.3 Pseudocode for one complete invocation

```
# online main loop
def turn(input):
    percept = perceive(input)                       # vision/speech/text
    entities = identify(percept)                    # person candidates + confidence
    sal = salience(percept, entities, goals)        # six-dimensional scoring
    event = encode(percept, entities, sal)          # event log + index
    cands = retrieve(percept.cues, entities, mod)   # pattern separation + completion
    rel = relationship(entities, cands)             # multi-dimensional + evidence
    packet = compile(cands, rel, goals, commitments, mod)
    packet = audit(packet)                          # sources + conflicts
    decision = main_brain.decide(packet)            # the only decision point
    result = effector.execute(decision)
    write_back(event, result)                       # RPE + state
    enqueue_consolidation(event, result)
    return decision.response
```

> CLOSING LINE
>
> This proposal is not meant to prove that "machines have memory," but to prove that **by treating memory as an auditable, prunable, reconstructable process, a machine can do better than "a vector store + a long prompt."**

APPENDIX D · REFERENCE IMPLEMENTATION

Reference Implementation

The code is not inside the HTML; it lives in `runtime/` and is verified with the standard-library unittest.

MODULE	RESPONSIBILITY	CORRESPONDING SECTION
`state_params.py`	Event state → bounded runtime parameters; hashing and bounds	2.7
`memory.py`	Retrieval, ContextPacket, omitted, budget	4.5 / 5.2 / 6.6
`ledger.py`	Evidence ledger, version chains, retention, cold archiving	4.6 / 6.3 / 6.4
`constitution.py`	Proposals, equal co-signature, cooling-off, rollback, ratification ledger	3.7
`segmentation.py`	Segmentation by prediction error / topic / person / gap / duration	5.1.1
`relationship.py`	Multi-dimensional relationships, affect/trust separation, interaction weights, idempotency	6.8

MODULE	RESPONSIBILITY	CORRESPONDING SECTION
`reliability.py`	Source types, verifiability, historical calibration, consistency	6.4
`simulation.py` / `primitives.py`	Compositional replay, primitive library, simulation writes no facts	6.2.1 / 6.7.2
`graph_retrieval.py`	Lightweight adjacency index, 1–2 hops, decay and budget	5.2
`pipeline.py`	End to end: event → parameters → packet → proposal → ratification → audit	Chapter 5

```
cd runtime
python3 -m unittest -v      # 128 tests
python3 pipeline.py         # end-to-end JSON + audit trace
```

Boundaries: this is an **in-memory reference implementation**, not a production system; it has no persistence, real signing keys, distributed writer, or production-grade ratification ledger. Its purpose is to validate contracts and failure modes, not to replace an engineering implementation.

DOUBLE-BRAIN SYSTEM • VERSION 2.1

This proposal is a mechanistic deepening and engineering recommendation draft of Dual-Brain System: A Multi-Model Collaborative Cognition Framework, Version 1. It retains the three core constraints of Version 1: there is only one integrating center; the Backend only processes information—it does not offer suggestions, act, or make decisions on its own; and knowledge, memory, and skills are managed separately. The neuroscience content in this document serves to explain design rationale; it does not constitute a complete model of the human brain, nor does it claim that the system possesses subjective consciousness.

Public version 2.1 · For research and engineering reference.